ELEMENTARY PROBABILITY

SECOND EDITION

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5
Random Vectors: Independence and Dependence

Wherever there is wealth there will be dependence and expectation.
Samuel Johnson [The Rambler, 189]

5.1 Joint Distributions

Commonly, each outcome of an experiment generates two (or more) real numbers of interest. We can treat these as individual random variables $(X_i; 1 \leq i \leq n)$, but it is often important to consider their joint behaviour. For example, if the experiment is your visit to your doctor, you may find out your height H and weight W. These are separate random variables, but are often informative when considered jointly. Thus, the outcome $H = 150\,\text{cm}$ and $W = 150\,\text{kg}$ might disturb your physician, whereas the outcome $H = 190\,\text{cm}$ and $W = 80\,\text{kg}$ probably would not. Likewise, the random vector comprising height, weight, age, sex, blood pressure, and heart rate is of more use considered jointly than separately.

As another example, complicated systems (e.g., space shuttles) have several on-board computers that work together to run the system. If one fails or makes an error, the others can override it; thus, the system fails only when a majority of the computers fail. If X_i is the time until the ith processor fails, then the time until the system fails depends jointly on the collection of random variables, $X_1, \ldots, X_n$. It is natural to refer to such a collection as a random vector, and write $X = (X_1, X_2, \ldots, X_n)$. Formally, as before, we have $X = X(\omega); \omega \in \Omega$ and $A_x = \{\omega : X(\omega) = x\} \in \mathcal{F}$, but we do not often refer to the underlying sample space Ω. Because X maps Ω into a countable subset S of R^n, we think of S as the sample space. (You may well have already been doing this instinctively in Chapter 4.)

For simplicity, we summarize the properties of random vectors in two dimensions; the appropriate generalizations in more dimensions are straightforward.

Definition *Let X and Y be two discrete random variables taking values $(x_i; i \geq 1)$ and $(y_j; i \geq 1)$, respectively. Their* joint probability mass function *$f(x, y)$ is defined by*

$$f(x, y) = \mathbf{P}(X = x, Y = y)$$

as x and y range over all possible values x_i and y_j of X and Y. ▲

The mass function $f(x, y)$ is zero, except at a countable set of points in $\mathbb{R}^2$. In fact,

$$f(x, y) \geq 0 \quad \text{for all } x \text{ and } y, \tag{1}$$

and further if $\sum_{i,j} f(x_i, y_j) = 1$, then the joint distribution $f(x, y)$ is not defective. Most important is the Key Rule:

$$\mathbf{P}((X, Y) \in C) = \sum_{(x,y)\in C} f(x, y). \tag{2}$$

(3) Example Suppose that a coin is tossed twice; let X be the total number of heads shown and Y the total number of tails. Then, (X, Y) takes values in

$$S = \{0, 1, 2\} \times \{0, 1, 2\} = \{(i, j) : i \in \{0, 1, 2\}, j \in \{0, 1, 2\}\}.$$

Clearly, $f(x, y)$ is zero, except at the points $(0, 2)$, $(1, 1)$, and $(2, 0)$. Furthermore,

$$f(0, 2) + f(1, 1) + f(2, 0) = (1 - p)^2 + 2p(1 - p) + p^2 = 1,$$

where we have denoted the probability of a head by p, as usual. ●

Any real function $g(X, Y)$ of two such jointly distributed random variables is itself a random variable. If we set $Z = g(X, Y)$, then Z has a probability mass function given by using the Key Rule (2) above:

$$f_Z(z) = \mathbf{P}(g(X, Y) = z) = \sum f(x, y), \tag{4}$$

where the summation is over all x and y, such that $g(x, y) = z$.

In particular, if $g(x, y) = x$, we have

$$f_X(x) = \sum_y f(x, y), \tag{5}$$

and if $g(x, y) = y$, we have

$$f_Y(y) = \sum_x f(x, y). \tag{6}$$

Thus, we have shown the important result that, if we know the joint mass function of several random variables, we can find all their separate mass functions. When obtained in this way, $f_X(x)$ and $f_Y(y)$ are sometimes called *marginal* mass functions.

Here are some examples illustrating joint mass functions.

(7) Example A row of n numbered machines are producing components that are identical, except for the serial number. On any day, the kth component produced by the jth machine bears the serial number (j, k). On the day in question, the rth machine produces c_r $(1 \leq r \leq n)$ components, and at the end of the day one component C is picked at random from all those produced. Let its serial number be (X, Y). Find $f(x, y)$, $f_X(x)$ and $f_Y(y)$.

Solution Because C is picked at random from all $\sum_{r=1}^{n} c_r$ components, we have

$$f(x, y) = \left(\sum_{1}^{n} c_r \right)^{-1}, \quad 1 \le x \le n; 1 \le y \le c_x,$$
$$= a \quad \text{(say)}.$$

Then, by (5),

$$f_X(x) = \sum_y f(x, y) = ac_x.$$

Now define the function $H(i, j) = \begin{cases} 1 & \text{if } c_i \ge c_j, \\ 0 & \text{otherwise.} \end{cases}$

Then, by (6),

$$f_Y(y) = \sum_x f(x, y) = a \sum_{x=1}^{n} H(x, y).$$

●

(8) Example: Cutting for the Deal It is customary, before engaging in a card game, to cut for the deal; each player removes a portion of the deck in turn, and then each reveals the bottom card of his segment. The highest card wins. For these to be random variables, we need to assign numerical values to the court cards, so we set J = 11, Q = 12, K = 13, A = 14, when aces are high.

(a) Art and Bart cut for deal, aces high. Let X be Art's card, and Y be Bart's card. Find the joint mass function of X and Y. Does it make any difference how many cards Art removes from the deck?
(b) Let V be the loser's card, and W the dealer's (winning) card. Find the joint mass function of V and W, and the separate mass functions of V and W.
(c) Find the mass function of the dealer's winning margin (namely, $W - V$).
(d) What is the mass function of the dealer's card when three players cut for deal?

Note that in the event of a tie, the deck is shuffled and the players cut again to choose the dealer.

Solution (a) Each random variable takes values in $\{2, 3 \ldots, 14\}$. Cutting the deck twice amounts to selecting two cards at random, and because ties are not allowed, $X \ne Y$. By symmetry, any two unequal values are equally likely to occur, so

$$f_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y) = \begin{Bmatrix} \frac{1}{12} \cdot \frac{1}{13} & x \ne y \\ 0 & x = y \end{Bmatrix} \quad 2 \le x, y \le 14.$$

It makes no difference how many cards Art removes with his cut.

(b) Of course, $W > V$, so

$$f_{V,W}(v, w) = \mathbf{P}(V = v, W = w) = \mathbf{P}(X = v, Y = w) + \mathbf{P}(X = w, Y = v)$$
$$= \begin{cases} \frac{2}{12} \cdot \frac{1}{13} = \frac{1}{78}; & 2 \le v < w \le 14 \\ 0 \cdot \qquad\quad ; & \text{otherwise.} \end{cases}$$

This is otherwise obvious, because the experiment amounts to choosing an unordered pair of unequal cards at random, with equal probability of choosing any pair. Hence, for $v < w$, $f(v, w) = \binom{13}{2}^{-1}$, as above. Now, by (5),

$$f_V(v) = \sum_{w=v+1}^{14} \frac{1}{78} = \frac{14 - v}{78}; \quad 2 \leq v \leq 13.$$

Then by (6),

$$f_W(w) = \sum_{2}^{w-1} \frac{1}{78} = \frac{w - 2}{78}; \quad 3 \leq w \leq 14.$$

(c) By (4),

$$f_Z(z) = \mathbf{P}(W - V = z) = \sum \frac{1}{78},$$

where the summation is over all v and w such that $w - v = z$. Because $z \leq v < w \leq 14$, there are exactly $13 - z$ terms in this sum, so

$$\mathbf{P}(W - V = z) = \frac{13 - z}{78}; \quad 1 \leq z \leq 12.$$

(d) Arguing as we did for (b), where now $u < v < w$, we have

$$\mathbf{P}(U = u, V = v, W = w) = \binom{13}{3}^{-1}.$$

Hence,

$$\begin{aligned} f_W(w) &= \sum_{2 \leq u < v < w} \binom{13}{3}^{-1}; \quad 4 \leq w \leq 14, \\ &= \sum_{3 \leq v < w} (v - 2)\binom{13}{3}^{-1} = \frac{1}{2}(w - 3)(w - 2)\binom{13}{3}^{-1}. \end{aligned}$$ ●

(9) Example For what value of c is the function

$$f(x, y) = c\binom{x + y - 1}{x}\lambda^x \mu^y, \quad x \geq 0; y \geq 1; 1 > 1 - \lambda > \mu > 0$$

a joint mass function? For this value of c, find the mass functions of X and Y.

Solution By (2),

$$\begin{aligned} c^{-1} &= \sum_{y=1}^{\infty} \sum_{x=0}^{\infty} \binom{x + y - 1}{x}\lambda^x \mu^y = \sum_{y=1}^{\infty} \frac{\mu^y}{(1 - \lambda)^y} \quad \text{by (3.6.12),} \\ &= \frac{\mu}{1 - \lambda - \mu}. \end{aligned}$$

Then, by (5),

$$f_X(x) = c\sum_{y=1}^{\infty} \mu\lambda^x \binom{x+1+y-2}{y-1}\mu^{y-1} = \frac{(1-\lambda-\mu)\lambda^x}{(1-\mu)^{x+1}}, \quad x \geq 0.$$

Likewise,

$$f_Y(y) = c\sum_{x=0}^{\infty} \binom{x+y-1}{x}\lambda^x\mu^y = \frac{(1-\lambda-\mu)\mu^{y-1}}{(1-\lambda)^y}, \quad y \geq 1.$$

Thus, $X+1$ and Y are both geometric, with parameters $\frac{\lambda}{1-\mu}$ and $\frac{\mu}{1-\lambda}$, respectively, X taking values in the nonnegative integers and Y in the positive integers. ●

(10) Example Leif and Rolf are bored with fair games. They want to play a game in which the probability of winning (for Leif) is λ, where λ is an arbitrary number in $[0,1]$. Also, they want the game to be of finite duration with probability 1. Unfortunately, the only gaming aid they have is a fair coin. Can you supply them with a game?

Solution Let λ have binary expansion

$$\lambda = 0.b_1b_2b_3\ldots = \sum_{n=1}^{\infty} b_n 2^{-n}.$$

Now toss the coin repeatedly and let I_n be the indicator of the event that the nth toss is a head. Let T be the first toss such that $I_n \neq b_n$, $T = \min\{n\colon I_n \neq b_n\}$. If $I_T < b_T$, then Leif wins; otherwise, Rolf wins. Now, $\mathbf{P}(T=n) = (\frac{1}{2})^n$ so that

$$\mathbf{P}(T<\infty) = \sum_{1}^{\infty}\left(\frac{1}{2}\right)^n = 1$$

and indeed $\mathbf{E}(T) = 2$. Also, Leif can only win at the nth toss if $b_n = 1$ so

$$\mathbf{P}(\text{Leif wins}) = \sum_n b_n\mathbf{P}(T=n) = \sum_n b_n 2^{-n} = \lambda,$$

as required. ●

5.2 Independence

Given the joint mass function of X and Y, equations (5.1.5) and (5.1.6) yield the marginal mass functions of X and Y. However, to be given the marginal distributions does not in general uniquely determine a joint distribution.

(1) Example Let X and Y have joint mass function given by

$$f(0,0) = \frac{1}{6},\ f(0,1) = \frac{1}{3},\ f(1,0) = \frac{1}{12},\ f(1,1) = \frac{5}{12},$$

and let U and V have joint mass function given by

$$f(0,0) = \frac{1}{4},\ f(0,1) = \frac{1}{4},\ f(1,0) = 0,\ f(1,1) = \frac{1}{2}.$$

Then, summing to get the marginal mass functions shows that:

$$f_X(0) = f_U(0) = \frac{1}{2},\ f_X(1) = f_U(1) = \frac{1}{2};$$

and

$$f_Y(0) = f_V(0) = \frac{1}{4},\ \ f_Y(1) = f_V(1) = \frac{3}{4}.$$

These marginal mass functions are the same, but the joint mass functions are different. ●

There is one exceptionally important special case when marginal mass functions do determine the joint mass function uniquely.

(2) Definition *Random variables X and Y are independent if, for all x and y,*

$$f(x, y) = f_X(x) f_Y(y).$$

This is equivalent to $\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$, where $A = \{\omega\colon X(\omega) = x\}$ and $B = \{\omega\colon X(\omega) = y\}$, which is the definition of independence for the events A and B. More generally, a collection $(X_i; 1 \le i \le n)$ with mass function f is independent if for all $x = (x_1, \ldots, x_n)$

(3)
$$f(x) = \prod_{i=1}^{n} f_{X_i}(x_i).$$

▲

Note that if X or Y (or both) are improper random variables [so that $f(x, y)$ is defective], then to say they are independent is interpreted as meaning

(4)
$$\mathbf{P}(X = x, Y = y) = \mathbf{P}(X = x)\mathbf{P}(Y = y)$$

for all finite x and y. This may seem odd, but such random variables occur quite naturally in simple random walks and other topics.

Example 5.1.7 Revisited Recall that n machines produce components. Suppose that all the machines produce the same number c of components, and as before we pick one at random and let its serial number be (X, Y), where X is the machine number and Y is the component index. Then,

$$f(x, y) = (nc)^{-1}; \quad 1 \le x \le n;\ 1 \le y \le c$$

and

$$f_X(x) = \frac{1}{n}; \quad 1 \le x \le n \qquad \text{and} \qquad f_Y(y) = \frac{1}{c}; \quad 1 \le y \le c.$$

Obviously, $f(x, y) = f_X(x) f_Y(y)$ and so X and Y are independent in this case. ●

(5) Example 1 Revisited Observe that the mass functions of X and Y, and of U and V, do not satisfy Definition 2. Let W and Z be independent random variables such that $f_W(0) = \frac{1}{2}$, $f_W(1) = \frac{1}{2}$ and $f_Z(0) = \frac{1}{4}$, $f_Z(1) = \frac{3}{4}$.

Then, by Definition 2, their joint mass function is

$$f(0,0) = \frac{1}{8},\ f(0,1) = \frac{3}{8},\ f(1,0) = \frac{1}{8},\ f(1,1) = \frac{3}{8}. \quad \bullet$$

(6) Example 5.1.9 Revisited Observe that X and Y are not independent because $\binom{x+y-1}{x}\lambda^x\mu^y$ cannot be expressed in the form $f_X(x)f_Y(y)$. If $X + 1$ and Y were independent geometric random variables with parameters $\frac{\lambda}{1-\mu}$ and $\frac{\mu}{1-\lambda}$, then the joint mass function would be

$$f(x,y) = \left(1 - \frac{\lambda}{1-\mu}\right)\left(1 - \frac{\mu}{1-\lambda}\right)\left(\frac{\lambda}{1-\mu}\right)^x\left(\frac{\mu}{1-\lambda}\right)^{y-1}, \quad x \geq 0; y \geq 1.$$

$\bullet$

The apparently simple Definition 2 implies a great deal more about independent random variables, as the following result shows.

(7) Theorem *Let X and Y be independent random variables. Then*:
(a) *For arbitrary countable sets A and B,*

(8) $$\mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B),$$

and
(b) *For any real functions $g(\cdot)$ and $h(\cdot)$, $g(X)$ and $h(Y)$ are independent.*

Proof (a) The left-hand side of (8) is

$$\begin{aligned}\sum_{x\in A}\sum_{y\in B}\mathbf{P}(X = x, Y = y) &= \sum_{x\in A}\sum_{y\in B}\mathbf{P}(X = x)\mathbf{P}(Y = y) \quad \text{by independence} \\ &= \sum_{x\in A}\mathbf{P}(X = x)\sum_{y\in B}\mathbf{P}(Y = y) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B),\end{aligned}$$

as required. For (b), let $A = \{x\colon g(X) = \xi\}$ and $B = \{y\colon h(Y) = \eta\}$. Then, by part (a), for any ξ and η,

$$\begin{aligned}\mathbf{P}(g(X) = \xi, h(Y) = \eta) &= \mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B) \\ &= \mathbf{P}(g(X) = \xi)\mathbf{P}(h(Y) = \eta),\end{aligned}$$

as required. ■

(9) Example Independent random variables X and Y take the values -1 or $+1$ only, and $\mathbf{P}(X = 1) = a$, $\mathbf{P}(Y = 1) = \alpha$. A third random variable Z is defined by $Z = \cos((X + Y)\frac{\pi}{2})$. If $0 < a, \alpha < 1$, show that there are unique values of a and α such that X and Z are independent, and Y and Z are independent. In this case, are X, Y, and Z independent?

Solution First, for Z,

$$\mathbf{P}(Z = 1) = \mathbf{P}(X + Y = 0) = a(1 - \alpha) + \alpha(1 - a)$$

and, likewise, $\mathbf{P}(Z = -1) = a\alpha + (1 - a)(1 - \alpha)$. Now,

$$\mathbf{P}(Z = 1, X = 1) = \mathbf{P}(X = 1, Y = -1) = a(1 - \alpha).$$

Hence, if

$$a(1 - \alpha) = a(a(1 - \alpha) + \alpha(1 - a)) \tag{10}$$

we have $\mathbf{P}(Z = 1, X = 1) = \mathbf{P}(Z = 1)\mathbf{P}(X = 1)$. Simplifying (10) yields $\alpha = \frac{1}{2}$. Now, plodding through three similar constraints shows that X and Z are independent iff $\alpha = a = \frac{1}{2}$. By symmetry, the same condition holds iff Y and Z are independent.

However, X, Y, and Z are not independent because

$$\mathbf{P}(X = 1, Y = 1, Z = -1) = 0 \neq \mathbf{P}(X = 1)\mathbf{P}(Y = 1)\mathbf{P}(Z = -1). \qquad \bullet$$

Independent random variables often have interesting and useful properties.

(11) Example Let X and Y be independent geometric random variables having respective mass functions $f_X(x) = (1 - \lambda)\lambda^x$ and $f_Y(y) = (1 - \mu)\mu^y$ for $x \geq 0$ and $y \geq 0$. What is the mass function of $Z = \min\{X, Y\}$?

Solution By independence,

$$\mathbf{P}(Z > n) = \mathbf{P}(X > n \cap Y > n) = \mathbf{P}(X > n)\mathbf{P}(Y > n) = \lambda^{n+1}\mu^{n+1} = (\lambda\mu)^{n+1}.$$

Hence,

$$\mathbf{P}(Z = n) = \mathbf{P}(Z > n - 1) - \mathbf{P}(Z > n) = (1 - \lambda\mu)(\lambda\mu)^n$$

and Z is also geometric with parameter $\lambda\mu$. $\bullet$

5.3 Expectation

Let the random variable $Z = g(X, Y)$ be a function of X and Y. Using (5.1.4) and the definition of expectation (4.3.1), we have

$$\mathbf{E}(Z) = \sum_z z f_Z(z) = \sum_z z\mathbf{P}(g(X, Y) = z).$$

This expression for $\mathbf{E}(Z)$ is not always simple or convenient for use in calculation. The following generalization of Theorem 4.3.4 is therefore very useful.

(1) Theorem *Let X and Y have joint mass function $f(x, y)$. Whenever the sum on the right-hand side is absolutely convergent, we have*

$$\mathbf{E}(g(X, Y)) = \sum_{x,y} g(x, y) f(x, y).$$

Proof The proof is essentially the same as that of Theorem 4.3.4. $\blacksquare$

(2) **Corollary** *For any real numbers a and b,*

$$\mathbf{E}(aX + bY) = a\mathbf{E}(X) + b\mathbf{E}(Y)$$

when both sides exist and are finite.

Proof Because the sum is absolutely convergent, by (1),

$$\begin{aligned}\mathbf{E}(aX + bY) &= \sum_{x,y}(ax + by)f(x, y) = \sum_{x,y} axf(x, y) + \sum_{x,y} byf(x, y)\\ &= \sum_{x} axf_X(x) + \sum_{y} byf_Y(y) \qquad \text{by (5.1.5) and (5.1.6)}\\ &= a\mathbf{E}(X) + b\mathbf{E}(Y).\end{aligned}$$

■

(3) **Example: Coupons** Recall Example 4.3.15 in which you were collecting coupons; we can now find $\mathbf{E}(R)$ more quickly. Let T_1 be the number of packets required to obtain the first coupon, T_2 the *further* number of packets required to obtain a second type of coupon, T_3 the *further* number required for the third type and so on. Then,

$$R = \sum_{k=1}^{n} T_k.$$

Obviously, $T_1 = 1$. Also,

$$\mathbf{P}(T_2 = r) = \left(\frac{1}{n}\right)^{r-1}\left(1 - \frac{1}{n}\right)$$

so that T_2 is geometric with mean $\frac{n}{n-1}$. Likewise, T_k is geometric with mean

$$\mathbf{E}(T_k) = \frac{n}{n - k + 1}; \quad 1 \le k \le n.$$

Hence, by (2),

$$\mathbf{E}(R) = \sum_{k=1}^{n} \mathbf{E}(T_k) = \sum_{k=1}^{n} \frac{n}{n - k + 1},$$

the same as the answer obtained with somewhat more effort in Example 4.3.15. ●

Corollary (2) is often useful when considering sums of indicators. For example, let $\{A_1, A_2, \ldots, A_n\}$ be any collection of events, and let

$$I_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{if } A_i \text{ does not occur} \end{cases}$$

be the indicator of A_i. Now, let X be the number of the A_i that occur. Then, by construction $X = \sum_{i=1}^{n} I_i$, and by (2)

$$\mathbf{E}(X) = \sum_{i=1}^{n} \mathbf{E}(I_i) = \sum_{i=1}^{n} \mathbf{P}(A_i).$$

We use this result in the following example.

Example: Binomial Distribution Let X be binomial with parameters n and p. Show that the factorial moments of X are given by

$$\mu^{(k)} = p^k n(n-1)\dots(n-k+1).$$

Solution Suppose a coin that shows heads with probability p is tossed n times. Then, the number of heads has the mass function of X. Let Y be the number of distinct sets of k such that all k tosses show heads. Then, $Y = \binom{X}{k}$. However, each of the $\binom{n}{k}$ distinct sets of k tosses shows k heads with probability p^k. Hence, $\mathbf{E}(Y) = \binom{n}{k}p^k$. Therefore, we have

$$\mathbf{E}\left(\binom{X}{k}\right) = \binom{n}{k}p^k,$$

which is the desired result. ●

We single certain expectations out for special notice. Just as random variables have moments, jointly distributed random variables have joint moments.

(4) Definition *The joint moments of X and Y are*

$$\mu_{ij} = \mathbf{E}(X^i Y^j); \quad i, j \geq 1. \qquad ▲$$

(5) Definition *The* covariance *of X and Y is*

$$\operatorname{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}(X))(Y - \mathbf{E}(Y))] = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y). \qquad ▲$$

This is the most important of the *central joint moments*, which are

$$\sigma_{ij} = \mathbf{E}[(X - \mathbf{E}(X))^i (Y - \mathbf{E}(Y))^j]; \quad i, j \geq 1.$$

Here are two interesting properties of $\operatorname{cov}(X, Y)$.

(6) Theorem *For jointly distributed random variables X and Y, and constants a, b, c, d, we have*:

(i) $\operatorname{cov}(aX + b, cY + d) = ac \operatorname{cov}(X, Y)$
(ii) $\operatorname{var}(X + Y) = \operatorname{var}(X) + \operatorname{var}(Y) + 2 \operatorname{cov}(X, Y)$

Proof

$$\begin{aligned}\text{(i) } \operatorname{cov}(aX + b, cY + d) &= \mathbf{E}[(aX + b - a\mathbf{E}(X) - b)(cY + d - c\mathbf{E}(Y) - d)] \\ &= \mathbf{E}[ac(X - \mathbf{E}(X))(Y - \mathbf{E}(Y))] = ac \operatorname{cov}(X, Y)\end{aligned}$$

$$\begin{aligned}\text{(ii) } \operatorname{var}(X + Y) &= \mathbf{E}(X + Y - \mathbf{E}(X) - \mathbf{E}(Y))^2 \\ &= \mathbf{E}[(X - \mathbf{E}(X))^2 + (Y - \mathbf{E}(Y))^2 + 2(X - \mathbf{E}(X))(Y - \mathbf{E}(Y))],\end{aligned}$$

as required. ■

Let us find $\operatorname{cov}(X, Y)$ for the simple examples we have met above.

(1) Examples 5.2.1 and 5.2.5 Revisited Find the covariance for each of the three joint mass functions given in these two examples.

Solution In every case,

$$\mathbf{E}(XY) = 1^2 f(1,1)$$

and $\mathbf{E}(X) = f(1,0) + f(1,1)$, and $\mathbf{E}(Y) = f(0,1) + f(1,1)$.
Hence,

$$\operatorname{cov}(X,Y) = f(1,1) - (f(1,0) + f(1,1))(f(0,1) + f(1,1)).$$

Evaluating this in the three given instances shows that:

(i) $$\operatorname{cov}(X,Y) = \frac{5}{12} - \left(\frac{1}{3} + \frac{5}{12}\right)\left(\frac{1}{12} + \frac{5}{12}\right) = \frac{1}{24}$$

(ii) $$\operatorname{cov}(U,V) = \frac{1}{2} - \left(\frac{1}{4} + \frac{1}{2}\right)\cdot\frac{1}{2} = \frac{1}{8}$$

(iii) $$\operatorname{cov}(W,Z) = \frac{3}{8} - \left(\frac{3}{8} + \frac{3}{8}\right)\left(\frac{1}{8} + \frac{3}{8}\right) = 0.$$ ●

(7) Example 5.1.8 Revisited Recall that Art and Bart are cutting for the deal. Find cov (X, Y) and cov (V, W).

Solution

$$\mathbf{E}(X) = \mathbf{E}(Y) = \sum_{2}^{14} \frac{x}{13} = 8.$$

Also, using (1),

$$\begin{aligned}\mathbf{E}(XY) &= \sum_{2\le x\ne y\le 14} \frac{x}{12}\cdot\frac{y}{13} = \frac{1}{12}\cdot\frac{1}{13}\sum_{y}(105 - y - 1)y \\ &= \frac{1}{12}\cdot\frac{1}{13}(105\times 104 - 1118) = 64 - \frac{7}{6}.\end{aligned}$$

Hence, $\operatorname{cov}(X,Y) = -\frac{7}{6}$.

Likewise, using the expressions in Example 5.1.8 for the marginal mass functions of V and W, we find

$$\begin{aligned}\mathbf{E}(V) &= \sum_{v=2}^{13} \frac{v(14-v)}{78} = \frac{1}{78}\sum_{v=2}^{13}(13v - v(v-1)) \\ &= +\frac{1}{78}\sum_{v=2}^{13}\left(\frac{13}{2}[v(v+1) - v(v-1)]\right. \\ &\quad \left. -\frac{1}{3}[(v+1)v(v-1) - v(v-1)(v-2)]\right) \\ &= \frac{17}{3}\end{aligned}$$

after successive cancellation of the terms in the sum.

Similarly, we find $\mathbf{E}(W) = \frac{31}{3}$.
Now for all ω, $X(\omega)Y(\omega) = V(\omega)W(\omega)$, so $\mathbf{E}(VW) = \mathbf{E}(XY)$, and finally,

$$\operatorname{cov}(V, W) = 62 + \frac{5}{6} - \frac{17}{3} \cdot \frac{31}{3} = +\frac{77}{18}. \quad \bullet$$

Just as joint mass functions have a simple form when random variables are independent, so too do joint moments simplify.

(8) Theorem *If X and Y are independent random variables with finite expectations, then $\mathbf{E}(XY)$ exists, and $\mathbf{E}(XY) = \mathbf{E}(X)\mathbf{E}(Y)$. It follows that* $\operatorname{cov}(X, Y) = 0$ *in this case.*

Proof By independence and Theorem 5.3.1,

$$\mathbf{E}(|XY|) = \sum_{x,y} |xy| f_X(x) f_Y(y) = \sum_x |x| f_X(x) \sum_y |y| f_Y(y) = \mathbf{E}(|X|)\mathbf{E}(|Y|)$$

so $\mathbf{E}(|XY|) < \infty$. Thus, $\mathbf{E}(XY)$ exists, and the same argument shows that $\mathbf{E}(XY) = \mathbf{E}(X)\mathbf{E}(Y)$. ■

(9) Definition *If* $\operatorname{cov}(X, Y) = 0$, *then X and Y are said to be* uncorrelated. *If* $\mathbf{E}(XY) = 0$, *then X and Y are said to be* orthogonal. ▲

It follows that independent random variables are uncorrelated, but the converse is not true, as the following example shows.

(10) Example A random variable X is said to be *symmetric* if $\mathbf{P}(X = -x) = \mathbf{P}(X = x)$ for all x. Let X be symmetric with $\mathbf{E}(X^3) < \infty$, and let $Y = X^2$. Then, because X has an expectation it is zero, by symmetry, and

$$\mathbf{E}(XY) = \mathbf{E}(Y^3) = \sum_{x>0} x^3(f(x) - f(-x)) = 0 = \mathbf{E}(X)\mathbf{E}(Y).$$

Thus, $\operatorname{cov}(X, Y) = 0$, even though X and Y are not independent. In this case, X and Y are uncorrelated and orthogonal, but dependent. ●

Thus, up to a point, and in a way that we carefully leave unspecified, $\operatorname{cov}(X, Y)$ can be an indication of the dependence of X and Y. It has the drawback that it depends on the scale of X and Y. Thus, if a is a constant, aX and Y have the same "dependence" as X and Y

(whatever we mean by that), but $\operatorname{cov}(aX, Y) = a \operatorname{cov}(X, Y)$. For this reason, statisticians more commonly use the following.

(11) Definition *The* correlation coefficient *of random variables X and Y is*

$$\rho(X, Y) = \frac{\operatorname{cov}(X, Y)}{(\operatorname{var}(X) \operatorname{var}(Y))^{\frac{1}{2}}},$$

whenever the right-hand side exists. ▲

Example 5.1.3 Revisited Here, X and Y are the number of heads and tails, respectively, when a coin is tossed twice. What are $\operatorname{cov}(X, Y)$ and $\rho(X, Y)$?

Solution Trivially,

$$\mathbf{E}(XY) = 1^2\mathbf{P}(X = 1, \mathbf{Y} = 1) = 2p(1 - p).$$

Likewise, $\mathbf{E}(X) = 2p$, $\mathbf{E}(Y) = 2(1 - p)$, $\operatorname{var}(X) = 2p(1 - p)$ and $\operatorname{var}(Y) = 2p(1 - p)$. Hence,

$$\operatorname{cov}(X, Y) = 2p(1 - p) - 4p(1 - p) = -2p(1 - p)$$

and

$$\rho(X, Y) = \frac{-2p(1 - p)}{(4p^2(1 - p)^2)^{\frac{1}{2}}} = -1.$$ ●

The correlation coefficient ρ has the following interesting properties; we assume that X and Y are not constant, and have finite variance.

(12) Theorem *If X and Y have correlation $\rho(X, Y)$, then*:

(i) $-1 \leq \rho(X, Y) \leq 1$.
(ii) $|\rho| = 1$ *if and only if* $\mathbf{P}(X = aY) = 1$ *for some constant a.*
(iii) $\rho(aX + b, cY + d) = \operatorname{sgn}(ac)\rho(X, Y)$, *where* $\operatorname{sgn}(x)$ *denotes the sign of x.*
(iv) $\rho = 0$ *if X and Y are independent.*

The proof of this theorem relies on the following important and useful result.

(13) Lemma: Cauchy–Schwarz Inequality *If* $\mathbf{E}(X^2)\mathbf{E}(Y^2) < \infty$, *then*

$$(\mathbf{E}(XY))^2 \leq \mathbf{E}(X^2)\mathbf{E}(Y^2). \tag{14}$$

Proof Suppose $0 < \mathbf{E}(X^2)\mathbf{E}(Y^2)$. By Theorem 4.3.6 (iii),

$$\begin{aligned} 0 &\leq \mathbf{E}[(X\mathbf{E}(Y^2) - Y\mathbf{E}(XY))^2] \\ &= \mathbf{E}(X^2)(\mathbf{E}(Y^2))^2 - 2\mathbf{E}(XY)^2\mathbf{E}(Y^2) + \mathbf{E}(Y^2)[\mathbf{E}(XY)]^2 \\ &= \mathbf{E}(Y^2)[\mathbf{E}(X^2)\mathbf{E}(Y^2) - (\mathbf{E}(XY))^2]. \end{aligned} \tag{15}$$

Because $\mathbf{E}(Y^2) > 0$, (14) follows. Of course, (14) is trivially true if $\mathbf{E}(Y^2) = 0$, for then $Y = XY = 0$ with probability one. ■

Proof of (12) (i) Applying Lemma 13 to the random variables $X - \mathbf{E}(X)$ and $Y - \mathbf{E}(Y)$ shows that $(\rho(X, Y))^2 \leq 1$, and so $-1 \leq \rho \leq 1$, as required.

(ii) If $|\rho| = 1$, then from (15),

$$\mathbf{E}[(X\mathbf{E}(Y^2) - Y\mathbf{E}(XY))^2] = 0,$$

and so from Example 4.6.10, with probability one $X = (\mathbf{E}(XY)/\mathbf{E}(Y^2))Y$.

(iii) Expanding, and using Theorem 6(i),

$$\rho(aX + b, cY + d) = \frac{ac \operatorname{cov}(X, Y)}{(a^2 \operatorname{var}(X) c^2 \operatorname{var}(Y))^{\frac{1}{2}}} = \frac{ac}{\sqrt{(ac)^2}} \rho(X, Y),$$

as required

(iv) This follows immediately from Theorem 8. ■

(16) Example (5.1.9) Revisited Recall that X and Y have joint mass function

$$f(x, y) = \frac{1 - \lambda - \mu}{\mu} \binom{x + y - 1}{x} \lambda^x \mu^y.$$

Show that $\rho(X, Y) = \left(\frac{\lambda\mu}{(1 - \lambda)(1 - \mu)} \right)^{\frac{1}{2}}$.

Solution First, we calculate $\mathbf{E}(XY)$ as

$$\begin{aligned} \frac{1 - \lambda - \mu}{\mu} \sum_{x,y} xy \binom{x + y - 1}{x} \lambda^x \mu^y &= \frac{1 - \lambda - \mu}{\mu} \sum_{y=1}^{\infty} y^2 \mu^y \lambda \\ \times \sum_{x=1}^{\infty} \binom{x + y - 1}{x - 1} \lambda^{x-1} &= \frac{(1 - \lambda - \mu)\lambda}{(1 - \lambda)\mu} \sum_{y=1}^{\infty} y^2 \left(\frac{\mu}{1 - \lambda} \right)^y \\ = \frac{\lambda(1 - \lambda + \mu)}{(1 - \lambda - \mu)^2}. \end{aligned}$$

Now we have already discovered in Example 5.1.9 that X and Y have geometric mass functions, so by Example 4.3.13

$$\begin{aligned} \mathbf{E}(X) &= \frac{1 - \mu}{1 - \lambda - \mu} - 1, \quad \mathbf{E}(Y) = \frac{1 - \lambda}{1 - \lambda - \mu} \\ \operatorname{var}(X) &= \frac{\mu(1 - \lambda)}{(1 - \lambda - \mu)^2}, \quad \operatorname{var}(Y) = \frac{\lambda(1 - \mu)}{(1 - \lambda - \mu)^2} \end{aligned}$$

and plugging all this into (11) yields

$$\rho = \frac{\lambda(1 - \lambda + \mu) - \lambda(1 - \lambda)}{(\mu(1 - \lambda)\lambda(1 - \mu))^{\frac{1}{2}}},$$

as required. ●

Finally, we remark that $\operatorname{cov}(X, Y)$ and $\rho(X, Y)$ are not the only functions used to measure dependence between X and Y. Another such function is

$$I(X, Y) = \sum_x \sum_y f(x, y) \log\left(\frac{f(x, y)}{f_X(x) f_Y(y)}\right) = \mathbf{E}\left(\log\left(\frac{f(X, Y)}{f_X(X) f_Y(Y)}\right)\right). \tag{17}$$

See Example 5.16 for more on this.

5.4 Sums and Products of Random Variables: Inequalities

These arise in many ways. For example, it is often useful to write a random variable as a sum of simpler random variables.

(1) Example: Binomial Random Variable The random variable X with mass function

$$f_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

has arisen in many ways; classically, it is the number of heads in n tosses of a biased coin. We now see that we can think about X in a different way. Let I_k be the indicator of the event that the kth toss of the coin shows a head. Then,

$$X = I_1 + I_2 + \cdots + I_n = \sum_{k=1}^{n} I_k.$$

We have written X as a sum of Bernoulli trials or indicators. Hence,

$$\mathbf{E}(X) = \mathbf{E}\left(\sum_{k=1}^{n} I_k\right) = \sum_{k=1}^{n} \mathbf{E}(I_k) = np.$$

Likewise,

$$\mathbf{E}(X^2) = \mathbf{E}\left(\sum_{k=1}^{n} I_k\right)^2 = \sum_{k=1}^{n} \mathbf{E}\left(I_k^2\right) + \sum_{j \neq k} \mathbf{E}(I_j I_k) = np + n(n-1)p^2.$$

Hence, $\operatorname{var}(X) = np(1 - p)$. You should compare this with your earlier methods using

$$\mathbf{E}(X^2) = \sum_{k=1}^{n} k^2 \binom{n}{k} p^k (1 - p)^{n-k} = \sum_{k=1}^{n} (k(k-1) + k) \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$

and so on. ●

(2) Theorem *Any discrete random variable X can be written as a linear combination of indicator random variables*; *thus*,

$$X = \sum_i a_i I(A_i)$$

for some collection of events $(A_i; i \geq 1)$ and real numbers $(a_i; i \geq 1)$.

Proof Just let $(a_i; i \geq 1)$ include the set of possible values of X, and set $A_i = \{\omega: X(\omega) = a_i\}$. ■

(3) Example: Matching Suppose that n distinct numbered keys ordinarily hang on n hooks bearing corresponding distinct numbers. On one occasion an inebriated turnkey hangs the keys at random on the hooks (one to each hook). Let X be the number of keys, which are then on the correct hooks. Find $\mathbf{E}(X)$ and var (X).

Solution Let I_j be the indicator of the event that the jth key does hang on the jth hook. Then,

$$X = \sum_{j=1}^{n} I_j.$$

Now by symmetry $\mathbf{P}(I_j = 1) = 1/n$ and for $j \neq k$,

$$\mathbf{P}(I_j I_k = 1) = \frac{1}{n(n-1)}. \tag{4}$$

Hence,

$$\mathbf{E}(X) = \mathbf{E}\left(\sum_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} \mathbf{E}(I_j) = \sum_{j=1}^{n} \mathbf{P}(I_j = 1) = 1.$$

Also,

$$\mathbf{E}(X^2) = \mathbf{E}\left(\sum_{j=1}^{n} I_j^2 + \sum_{j \neq k} I_j I_k\right) = 1 + 1,$$

using (4), and the fact that $I_j^2 = I_j$. Hence, var $(X) = 1$. ●

Indicators can also be useful when multiplied together; here is an illustration.

(5) Example Let us prove (1.4.8). Recall that we have events $A_1, \ldots, A_n$, and we seek the probability that at least one of them occurs, namely,

$$\mathbf{P}\left(\bigcup_{j=1}^{n} A_j\right) = t_n \quad \text{(say)}.$$

For economy of notation, we set

$$s_r = \sum_{i_1 < i_2 < \cdots < i_r} \mathbf{P}\left(A_{i_1} \cap \ldots \cap A_{i_r}\right); \quad 1 \leq r \leq n. \tag{6}$$

We let I_j be the indicator of the event that A_j occurs, and set

$$S_r = \sum_{i_1 < i_2 < \cdots < i_r} I_{i_1} I_{i_2} \ldots I_{i_r}.$$

Next, observe that

$$1-\prod_{j=1}^{n}(1-I_j)=\begin{cases}1 & \text{if at least one } A_j \text{ occurs}\\ 0 & \text{otherwise}\end{cases} \tag{7}$$

Hence, this is the indicator of the event whose probability we seek, and

$$\begin{aligned}\mathbf{P}\left(\bigcup_{j=1}^{n}A_j\right)&=\mathbf{E}\left(1-\prod_{j=1}^{n}(1-I_j)\right)\\ &=\mathbf{E}(S_1-S_2+\cdots+(-)^{n+1}S_n) \quad \text{on multiplying out,}\\ &=s_1-s_2+\cdots+(-)^{n+1}s_n \qquad \text{by (6), as required.}\end{aligned}$$ ●

The same expansion (7) can be used to prove the following interesting inequalities.

TheoremTheorem: Inclusion–Exclusion Inequalities *With the notation of Example* 5, *for* $1\le r\le n$,

$$(-)^r\left(\mathbf{P}\left(\bigcup_{j=1}^{n}A_j\right)-s_1+s_2-\cdots(-)^r s_r\right)\ge 0.$$

Proof First, we prove a simple identity. Obviously, $(1+x)^k(1+\frac{1}{x})^{-1}=x(1+x)^{k-1}$. Hence, equating the coefficient of x^{r+1} on each side gives

$$\binom{k}{r+1}-\binom{k}{r+2}+\cdots+(-)^{k-r+1}\binom{k}{k}=\binom{k-1}{r}\ge 0. \tag{8}$$

Furthermore, we have

$$\begin{aligned}&1-\prod_{1}^{n}(1-I_j)-S_1+S_2-\cdots+(-)^r S_r\\ &\quad=(-)^r(S_{r+1}-S_{r+2}+\cdots+(-)^{n-r+1}S_n).\end{aligned} \tag{9}$$

Now suppose exactly k of $A_1,\ldots,A_n$ occur. If $k\le r$, the right-hand side of (9) is zero. If $k>r$, the contribution in the bracket on the right-hand side is

$$\begin{aligned}\binom{k}{r+1}-\binom{k}{r+2}+\cdots+(-)^{k-r+1}\binom{k}{k}&=\binom{k-1}{r} \quad \text{by (8),}\\ &>0.\end{aligned}$$

Hence, no matter how many A_js occur

$$\begin{aligned}&(-)^r\mathbf{E}\left(1-\prod_{1}^{n}(1-I_j)-S_1+S_2+\cdots+(-)^r S_r\right)\\ &\quad=(-)^{2r}\mathbf{E}(S_{r+1}-\cdots+(-)^{t-r+1}S_n)\ge 0,\end{aligned}$$

as required. ■

It can similarly be shown (or deduced from the above) that if

$$t_r = \sum_{i_1<i_2<\cdots<i_r} \mathbf{P}\big(A_{i_1} \cup \cdots \cup A_{i_r}\big),$$

then $t_n = \sum_{r=1}^{n} (-)^{r-1} s_r$, and

$$(-)^r \left(\mathbf{P}\left(\bigcap_1^n A_i \right) - t_1 + t_2 - \cdots + (-)^r t_r \right) \geq 0.$$

Often, it is natural and important to consider the sum Z of two variables X and Y. This is itself a random variable, and so we may require the distribution of $Z = X + Y$. This is given by Example 5.1.3, so we have proved that

$$f_Z(z) = \sum_x f_{X,Y}(x, z - x). \tag{10}$$

One special case of this result must be singled out.

(11) Theorem *If X and Y are independent discrete random variables, then $Z = X + Y$ has probability mass function*

$$f_Z(z) = \sum_x f_X(x) f_Y(z - x). \tag{12}$$

Proof Substitute Definition 5.2.2 into (10). A summation of this form is called a *convolution*. ■

(13) Example: Sum of Geometric Random Variables (a) Let X_1 and X_2 be independent random variables, each having a geometric distribution

$$\mathbf{P}(X_i = k) = (1 - p)^k p; \quad k \geq 0. \tag{14}$$

Show that the mass function of $X_1 + X_2$ is $f(z) = (z + 1)(1 - p)^z p^2; z \geq 0$.

(b) If $(X_i; i > 1)$ are independent random variables, each having the geometric distribution (14), find the mass function of $Z = \sum_1^n X_i$.

Solution (a) Using (12),

$$f(z) = \sum_{k=0}^{z} (1 - p)^k p (1 - p)^{z-k} p = \sum_{k=0}^{z} p^2 (1 - p)^z = (z + 1) p^2 (1 - p)^z.$$

Alternatively, suppose we have a coin that, when tossed, shows a head with probability p. Toss this coin repeatedly until a head first appears, and let T_1 be the number of tails shown up to that point. Continue tossing the coin until the second head appears and let T_2 be the further number of tails shown up to that point. Then, T_1 and T_2 are independent and have the geometric distribution (14), and $T_1 + T_2$ is the number of tails shown before the second head appears. But we know from Example 4.2.8 that

$$\mathbf{P}(T_1 + T_2 = z) = \binom{z + 1}{1} p^2 (1 - p)^z.$$

Thus, $X_1 + X_2$ also has this mass function.

(b) Extending the second argument above, we see that $\sum_1^n X_i$ has the same distribution as the number of tails shown upto the point where n heads have first appeared in successive tosses of our biased coin. But the probability that z tails have appeared is just

$$\binom{z+n-1}{n-1}p^n(1-p)^z \tag{15}$$

and so this is the mass function of Z. It is negative binomial.

Now that we are in possession of the answer, we can use (12) to verify it by induction. Assume that $\sum_1^n X_i$ has mass function (15). Then, by (12), the mass function of $\sum_1^{n+1} X_i$ is

$$\begin{aligned}\mathbf{P}\left(\sum_{i=1}^{n+1} X_i = z\right) &= \sum_{k=0}^{z} p(1-p)^k\binom{z+n-k-1}{n-1}(1-p)^{z-k}p^n \\ &= \sum_{k=0}^{z} p^{n+1}(1-p)^z\binom{z+n-k-1}{n-1} = p^{n+1}(1-p)^z\binom{z+n}{k},\end{aligned}$$

as required, where we have used the identity

$$\sum_{r=0}^{z}\binom{n-1+r}{n-1} = \binom{n+z}{n}. \tag{16}$$

Because (15) holds for $n = 1$, the result follows by induction. Note that (16) can be derived immediately by equating the coefficient of x^z on both sides of the trivial identity

$$\frac{1}{1+x}\cdot\frac{1}{(1+x)^n} = \frac{1}{(1+x)^{n+1}}.$$

●

Example: Sum of Binomial Random Variables Let X and Y be independent random variables with respective mass functions

$$f_X(x) = \binom{m}{x}p^x(1-p)^{m-x}, \quad \text{and} \quad f_Y(y) = \binom{n}{y}p^y(1-p)^{n-y}.$$

Show that $X + Y$ has a binomial distribution.

Solution The expeditious method of doing this is to use Example 5.4.1 to write $X = \sum_{i=1}^m I_i$ and $Y = \sum_{i=m+1}^{m+n} I_i$. Hence,

$$X + Y = \sum_{i=1}^{m+n} I_i;$$

this has the B$(m + n, p)$ mass function by Example 5.4.1.

Alternatively, this may be shown by using (5.4.12); this is an exercise for you. ●

Turning to expected values, we recall that from Definition 5.2.2, for any random variables $(X_i; i \geq 1)$ with finite expectation, *whether they are independent or not*,

$$\mathbf{E}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\mathbf{E}(X_i). \tag{17}$$

If in addition the X_i are independent, then

$$\mathbf{E}\left(\prod_{i=1}^{n} X_i\right) = \prod_{i=1}^{n} \mathbf{E}(X_i). \tag{18}$$

Many inequalities for probabilities and moments are useful when dealing with random vectors. Most of them are beyond our scope at this stage, but we give a few simple examples with some applications.

(19) Basic Inequality If $X \leq Y$ with probability one, then $\mathbf{E}(X) \leq \mathbf{E}(Y)$.

Proof This follows immediately from Theorem 4.3.6. ■

Corollary *(a) For* $1 < r < s$,

$$\mathbf{E}(|X|^r) \leq (\mathbf{E}(|X|^s)) + 1. \tag{20}$$

(b) For $r \geq 1$,

$$\mathbf{E}(|X + Y|^r) \leq 2^r(\mathbf{E}(|X|^r) + \mathbf{E}(|Y|^r)). \tag{21}$$

Proof (a) If $|x| \leq 1$, then $|x|^r \leq 1$; and if $|x| > 1$ and $r < s$, then $|x|^r \leq |x|^s$. Hence, in any case, when $r \leq s$, $|x|^r \leq |x|^s + 1$. Thus,

$$\mathbf{E}(|X|^r) = \sum_x |x|^r f(x) \leq \sum_x |x|^s f(x) + 1 = \mathbf{E}(|X|^s) + 1.$$

(b) For any real numbers x and y, if $k \leq r$, $|x|^k|y|^{r-k} \leq |x^r + |y|^r$, because either $(|x|^k/|y|^k) \leq 1$ or $(|y|^{r-k}/|x|^{r-k}) < 1$. Hence,

$$\begin{aligned} |x + y|^r &\leq (|x| + |y|)^r \leq \sum_{k=0}^{r} \binom{r}{k} |x|^k|y|^{r-k} \\ &\leq \sum_{k=0}^{r} \binom{r}{k} (|x|^r + |y|^r) = 2^r(|x|^r + |y|^r), \end{aligned}$$

and (3) follows. ■

(22) Corollary *These inequalities show that:*

(a) *If* $\mathbf{E}(X^s) < \infty$, *then for all* $1 \leq r \leq s$, $\mathbf{E}(X^r) < \infty$.
(b) If $\mathbf{E}(X^r)$ *and* $\mathbf{E}(Y^r)$ *are finite then* $\mathbf{E}((X + Y)^r) < \infty$. ■

5.5 Dependence: Conditional Expectation

Let X and Y be jointly distributed random variables. We may be given the value of Y, either in fact, or as a supposition. What is the effect on the distribution of X?

(1) **Definition** *If X and Y have joint probability mass function $f(x, y)$, then given $Y = y$, the random variable X has a* conditional probability mass function *given by*

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(x)}$$

for all y such that $f_Y(y) > 0$. ▲

Example Let X and Y be independent geometric random variables each having mass function $f(x) = (1 - \lambda)\lambda^x; x \geq 0, 0 < \lambda < 1$. Let $Z = X + Y$. Show that for $0 \leq x \leq z$, $f_{X|Z}(x|z) = 1/(z + 1)$.

Solution From Example 5.4.13, we know that Z has mass function $f_Z(z) = (z + 1)(1 - \lambda)^2\lambda^z$ and so

$$f_{X|Z}(x|z) = \frac{\mathbf{P}(X = x, Z = z)}{(z + 1)(1 - \lambda)^2\lambda^z} = \frac{(1 - \lambda)^2\lambda^x\lambda^{z-x}}{(1 - \lambda)^2\lambda^z(z + 1)} = (z + 1)^{-1}.$$ ●

(2) **Example 5.1.8 Revisited: Cutting for the Deal** Find the conditional mass function of the loser's card conditional on $W = w$; find also $f_{W|V}(w|v)$.

Solution According to Example 5.1.8, $f(v, w) = \frac{1}{78}; 2 \leq v < w \leq 14$, and

$$f_W(w) = \frac{w - 2}{78}; \quad 3 \leq w \leq 14.$$

Hence, using Definition 1,

$$f_{V|W}(v|w) = \frac{1}{w - 2}; \quad 2 \leq v < w.$$

The loser's score is uniformly distributed given W. Likewise,

$$f_{W|V}(w|v) = \frac{1}{78} \Big/ \frac{14 - v}{78} = \frac{1}{14 - v}; \quad v < w \leq 14,$$

also a uniform distribution. ●

(3) **Example** Let X and Y be independent. Show that the conditional mass function of X given Y is $f_X(x)$, the marginal mass function.

Solution Because X and Y are independent, $f(x, y) = f_X(x)f_Y(y)$. Hence, applying Definition 1,

$$f_{X|Y}(x|y) = f(x, y)/f_Y(y) = f_X(x).$$ ●

(4) **Theorem** *$f_{X|Y}(x|y)$ is a probability mass function, which is to say that*

(i) $f_{X|Y}(x|y) \geq 0$ *and*
(ii) $\sum_x f_{X|Y}(x|y) = 1$.

Proof Part (i) is trivial. Part (ii) follows immediately from (5.1.5). ■

Recall that two events A and B are said to be conditionally independent given C, if $\mathbf{P}(A \cap B|C) = \mathbf{P}(A|C)\mathbf{P}(B|C)$. Likewise, it is possible for two random variables X and Y to be conditionally independent given Z, if

$$f_{X,Y|Z} = f_{X|Z} f_{Y|Z}.$$

Example: Cutting for the Deal (Example 5.1.8 Revisited) Suppose three players cut for the deal (with ties not allowed, as usual). Let X be the lowest card, Y the highest card and Z the intermediate card. Clearly, X and Y are dependent. However, conditional on $Z = z$, X and Y are independent. The mass function $f_{X|Z}$ is uniform on $\{2, \ldots, z-1\}$ and $f_{Y|Z}$ is uniform on $\{z+1, \ldots, 14\}$. ●

Being a mass function, $f_{X|Y}$ may have an expectation; it has a special name and importance.

(5) **Definition** *The* conditional expectation *of X, given that $Y = y$ where $f_Y(y) > 0$, is*

$$\mathbf{E}(X|\{Y = y\}) = \sum_x x f(x, y)/f_Y(y),$$

when the sum is absolutely convergent. ▲

As y varies over the possible values of Y, this defines a function of Y, denoted by $\mathbf{E}(X|Y)$. Because it is a function of Y, it is itself a random variable, which may have an expectation.

(6) **Theorem** *If both sides exist,*

(7) $$\mathbf{E}(\mathbf{E}(X|Y)) = \mathbf{E}(X).$$

Proof Assuming the sums are absolutely convergent we have, by Theorem 4.3.4,

$$\begin{aligned}\mathbf{E}(\mathbf{E}(X|Y)) &= \sum_y \mathbf{E}(X|\{Y = y\}) f_Y(y) = \sum_y \sum_x \frac{x f(x, y)}{f_Y(y)} f_Y(y) \quad \text{by Example 2} \\ &= \sum_x x f_X(x) \qquad \text{by (5.1.4)} \\ &= \mathbf{E}(X).\end{aligned}$$ ■

This is an exceptionally important and useful result. Judicious use of Theorem 6 can greatly simplify many calculations; we give some examples.

(8) Example: Eggs A hen lays X eggs where X is Poisson with parameter λ. Each hatches with probability p, independently of the others, yielding Y chicks. Show that $\rho(X, Y) = \sqrt{p}$.

Solution Conditional on $X = k$, the number of chicks is binomial $\mathrm{B}(k, p)$, with mean kp. Hence,

$$\mathbf{E}(XY) = \mathbf{E}(\mathbf{E}(XY|X)) = \mathbf{E}(X^2 p) = (\lambda^2 + \lambda)p.$$

Likewise,

$$\begin{aligned}\mathbf{E}(Y^2) &= \mathbf{E}(\mathbf{E}(Y^2|X)) = \mathbf{E}(X_p(1-p) + X^2 p^2) \\ &= \lambda p(1-p) + (\lambda + \lambda^2)p^2 = \lambda p + \lambda^2 p^2.\end{aligned}$$

Hence,

$$\rho(X, Y) = \frac{\mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y)}{(\mathrm{var}\,(X)\,\mathrm{var}\,(Y))^{\frac{1}{2}}} = \frac{\lambda^2 p + \lambda p - \lambda.\lambda p}{(\lambda(\lambda p + \lambda^2 p^2 - \lambda^2 p^2))^{\frac{1}{2}}} = \sqrt{p}. \qquad \bullet$$

(9) Example: Variance of a Random Sum Let $X_1, X_2, \ldots$ be a collection of independent identically distributed random variables, and let Y be an integer valued random variable independent of all the X_i. Let $S_Y = \sum_{i=1}^{Y} X_i$. Show that

$$\mathrm{var}\,(S_Y) = \mathbf{E}\big(X_1\big)^2\,\mathrm{var}\,(Y) + \mathbf{E}(Y)\mathrm{var}\,(X_1).$$

Solution By (7),

$$\mathbf{E}(S_Y) = \mathbf{E}(\mathbf{E}(S_Y|Y)) = \mathbf{E}\left(\mathbf{E}\left(\sum_1^Y X_i|Y\right)\right) = \mathbf{E}(Y\mathbf{E}(X_1)) = \mathbf{E}(Y)\mathbf{E}(X_1).$$

Likewise,

$$\begin{aligned}\mathbf{E}\big(S_Y^2\big) &= \mathbf{E}\big(\mathbf{E}\big(S_Y^2|Y\big)\big) = \mathbf{E}\big(Y\mathbf{E}\big(X_1^2\big) + Y(Y-1)[\mathbf{E}(X_1)]^2\big) \\ &= \mathbf{E}(Y)\big(\mathbf{E}\big(X_1^2\big) - [\mathbf{E}(X_1)]^2\big) + \mathbf{E}(Y^2)[\mathbf{E}(X_1)]^2,\end{aligned}$$

and so substituting into $\mathrm{var}\,(S_Y) = \mathbf{E}(S_Y^2) - (\mathbf{E}(S_Y))^2$, gives the result. $\bullet$

(10) Example 4.12 Revisited: Gamblers' Ruin Two gamblers, A and B, have n coins. They divide this hoard by tossing each coin; A gets those that show heads, X say, B gets the rest, totalling $n - X$.

They then play a series of independent fair games; each time A wins he gets a coin from B, each time he loses he gives a coin to B. They stop when one or other has all the coins.

Let D_X be the number of games played. Find $\mathbf{E}(D_X)$, and show that, when the coins are fair, $\rho(X, D_X) = 0$.

Solution Conditional on $X = k$, as in Example 4.12,

$$D_k = \frac{1}{2}D_{k+1} + \frac{1}{2}D_{k-1} + 1$$

with solution $D_k = k(n-k)$. Hence, observing that X is $B(n, p)$ (where p is the chance of a head), we have

$$\mathbf{E}(D_X) = \mathbf{E}(\mathbf{E}(D_X|X)) = \mathbf{E}(X(n-X)) = n(n-1)p(1-p).$$

Finally,

$$\text{cov}(X, D_X) = \mathbf{E}(X^2(n-X)) - \mathbf{E}X\mathbf{E}D_X = n(n-1)p(p-1)(2p-1),$$

whence $\rho = 0$, when $p = \frac{1}{2}$

(11) Example **Partition Rule:** Show that if X and Y are jointly distributed, then

$$f_X(x) = \sum_y f_Y(y) f_{X|Y}(x|y). \quad \bullet$$

Solution This is just Theorem 6 in the special case when we take X to be I_x, the indicator of the event $\{X = x\}$. Then, $\mathbf{E}(I_x) = f_X(x)$, and $\mathbf{E}(I_x|Y = y) = f_{X|Y}(x|y)$. The result follows from (7). Alternatively, you can substitute from Definition 1. Essentially this is the Partition Rule applied to discrete random variables. ●

Recall that we have already defined $\mathbf{E}(X|B)$ for any event B in Chapter 4. It is convenient occasionally to consider quantities, such as $\mathbf{E}(X|Y; B)$. This is defined to be the expected value of the conditional distribution

$$\mathbf{P}(X = x|\{Y = y\} \cap B) = \frac{\mathbf{P}(\{X = x\} \cap \{Y = y\} \cap B)}{\mathbf{P}(\{Y = y\} \cap B)} \tag{12}$$

for any value y of Y such that $\mathbf{P}(\{Y = y\} \cap B) > 0$.

We give some of the more important properties of conditional expectation.

(13) Theorem *Let a and b be constants, $g(.)$ an arbitrary function, and suppose that X, Y, and Z are jointly distributed. Then* (*assuming all the expectations exist*),

(i) $\mathbf{E}(a|Y) = a$
(ii) $\mathbf{E}(aX + bZ|Y) = a\mathbf{E}(X|Y) + b\mathbf{E}(Z|Y)$
(iii) $\mathbf{E}(X|Y) \geq 0$ if $X \geq 0$
(iv) $\mathbf{E}(X|Y) = \mathbf{E}(X)$, if X and Y are independent
(v) $\mathbf{E}(Xg(Y)|Y) = g(Y)\mathbf{E}(X|Y)$
(vi) $\mathbf{E}(X|Y; g(Y)) = \mathbf{E}(X|Y)$
(vii) $\mathbf{E}(\mathbf{E}(X|Y; Z)|Y) = \mathbf{E}(X|Y)$.

Property (v) is called the *pull-through property*, for obvious reasons.
Property (vii) is called the *tower property*. It enables us to consider multiple conditioning by taking the random variables in any convenient order.

Proof We prove the odd parts of Theorem 13, the even parts are left as exercises for you.

(i) $f(a, y) = f_Y(y)$, so

$$\mathbf{E}(a|Y) = af_Y(y)/f_Y(y) = a.$$

(iii) If $X \geq 0$, then every term in the sum in Theorem 6 is nonnegative. The result follows.

(v)
$$\mathbf{E}(Xg(Y)|Y = y) = \sum_x xg(y)f(x, y)/f_Y(y) = g(y)\sum_x f(x, y)/f_Y(y)$$
$$= g(y)\mathbf{E}(X|Y = y).$$

(vii) For arbitrary values, $Y = y$ and $Z = z$ of Y and Z, we have $\mathbf{E}(X|Y; Z) = \sum_x xf(x, y, z)/f_{Y,Z}(y, z)$. Hence, by definition,

(14)
$$\mathbf{E}(\mathbf{E}(X|Y; Z)|Y) = \sum_z \mathbf{E}(X|Y; Z)f_{Y,Z}(y, z)/f_Y(y) = \sum_z \sum_x xf(x, y, z)/f_Y(y)$$
$$= \sum_x xf(x, y)/f_Y(y) = \mathbf{E}(X|Y). \quad \blacksquare$$

(15) Example Three children (Aelhyde, Beowulf, and Canute) roll a die in the order $A, B, C, A, \ldots$, etc. until one of them rolls a six and wins. Find the expected number of rolls, given that Canute wins.

Solution We use a form of the tower property, (14). Let C be the event that Canute wins, let X be the duration of the game, and let Y denote the first roll to show a six in the first three rolls, with $Y = 0$ if there is no six. Then,

(16)
$$\mathbf{E}(X|C) = \mathbf{E}(\mathbf{E}(X|Y; C)|C).$$

Now if $Y = 0$, then with the fourth roll the game stands just as it did initially, except that three rolls have been made. So

$$\mathbf{E}(X|Y = 0; C) = 3 + \mathbf{E}(X|C).$$

Obviously, Y is otherwise 3, and

$$\mathbf{E}(X|Y = 3; C) = 3.$$

Therefore, substituting in (16), we have

$$\mathbf{E}(X|C) = 3 + \mathbf{E}(X|C)\mathbf{P}(Y = 0|C)$$

and so

$$\mathbf{E}(X|C) = \frac{3}{1 - \left(\frac{5}{6}\right)^3} = \frac{648}{91}.$$

Of course, there are other ways of doing this. ●

Finally, we remark that conditional expectation arises in another way.

(17) Theorem *Let $h(Y)$ be any function of Y such that $\mathbf{E}(h(Y)^2) < \infty$. Then,*

(18)
$$\mathbf{E}((X - h(Y))^2) \geq \mathbf{E}((X - \mathbf{E}(X|Y))^2).$$

Further, if $h(Y)$ is any function of Y such that

$$\mathbf{E}((X - h(Y))^2) = \mathbf{E}((X - \mathbf{E}(X|Y))^2), \tag{19}$$

then

$$\mathbf{E}((h(Y) - \mathbf{E}(X|Y))^2) = 0.$$

Proof

$$\begin{aligned} \mathbf{E}((X - h(Y))^2) &= \mathbf{E}((X - \mathbf{E}(X|Y) + \mathbf{E}(X|Y) - h(Y))^2) \\ &= \mathbf{E}((X - \mathbf{E}(X|Y))^2) + \mathbf{E}((\mathbf{E}(X|Y) - h(Y))^2) \\ &\quad + 2\mathbf{E}((X - \mathbf{E}(X|Y)) + (\mathbf{E}(X|Y) - h(Y))). \end{aligned} \tag{20}$$

However, by (7), we can write

$$\begin{aligned} \mathbf{E}((X - \mathbf{E}(X|Y))(\mathbf{E}(X|Y) - h(Y))) &= \mathbf{E}(\mathbf{E}((X - \mathbf{E}(X|Y))(\mathbf{E}(X|Y) - h(Y))|Y)) \\ &= \mathbf{E}(\mathbf{E}(X|Y) - h(Y))\mathbf{E}((X - \mathbf{E}(X|Y))|Y) \\ &\qquad \text{by Theorem 13(v)} \\ &= 0, \end{aligned}$$

because $\mathbf{E}((X - \mathbf{E}(X|Y))|Y) = 0$. The result (18) follows because $\mathbf{E}((\mathbf{E}(X|Y) - h(Y))^2) \geq 0$. Finally, if (19) holds then from (20), we have

$$\mathbf{E}((\mathbf{E}(X|Y) - h(Y))^2) = 0,$$

as required to complete the proof of Theorem 17. ■

Recall that if $\mathbf{E}(X^2) = 0$, then $\mathbf{P}(X = 0) = 1$; hence, if $\mathbf{E}((X - Y)^2) = 0$, then $X = Y$ with probability one. This suggests that the smaller $\mathbf{E}((X - Y)^2)$ is, then the "closer" X is to Y, in some sense. The point of the theorem is then that among all functions of Y, $\mathbf{E}(X|Y)$ is the one which is "closest" to X.

It is thus possible, and in later work desirable, to define $\mathbf{E}(X|Y)$ by this property. However, to explore all these ideas would take us too far afield.

5.6 Simple Random Walk

The ideas above now enable us to consider an exceptionally famous and entertaining collection of random variables.

(1) Definition *Let $(X_i; i \geq 1)$ be a collection of independent identically distributed random variables with mass function*

$$\mathbf{P}(X_1 = 1) = p; \mathbf{P}(X_1 = -1) = q = 1 - p.$$

Then the collection $(S_n; n \geq 0)$, where

$$S_n = S_0 + \sum_{i=1}^{n} X_i, \tag{2}$$

is called a simple random walk. *If $p = q$, it is called a* symmetric *simple random walk.* ▲

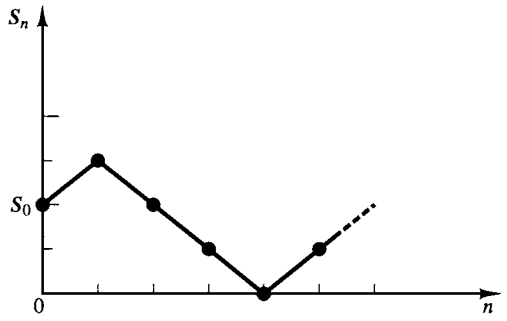


Figure 5.1 A path (or realization) of a simple random walk. The points represent successive positions (or values) of the walk; these are joined by steps of the walk. In this case, $S_0 = 2$, $S_1 = 3$, $S_2 = 2$, and so on. The walk visits the origin at the fourth step.

The nomenclature follows from the visualization of S_n as representing the position of a particle that is initially at S_0, and then takes a series of independent unit steps; each step being positive with probability p or negative with probability q.

It is conventional to display the walk in Cartesian coordinates as the sequence of points (n, S_n) for $n \geq 0$. Any particular such sequence is called a path of the random walk; see Figure 5.1 for an illustration.

Much of the effort of the first probabilists (Fermat, Pascal, Bernoulli, de Moivre, Laplace) was devoted to discovering the properties of the simple random walk, and more general random walks are still being investigated by modern probabilists.

It is easy to see that the celebrated gambler's ruin problem of Example 2.11 is just a simple random walk in which S_0 is interpreted as the initial capital of the gambler, and the walk stops on the first occasion D when either $S_n = 0$ (the gambler is ruined) or $S_n = K$ (his opponent is ruined). That is, the random variable

$$D = \min\{n : \{S_n = 0\} \cup \{S_n = K\}\} \tag{3}$$

is the duration of the game. In the context of random walks, D is called the first passage time of the walk to $\{0, K\}$.

The first thing to find is the mass function of S_n (when it is not stopped).

(4) Theorem

$$\mathbf{P}(S_n - S_0 = k) = \binom{n}{\frac{1}{2}(n+k)} p^{\frac{1}{2}(n+k)} q^{\frac{1}{2}(n-k)}.$$

Proof Consider a path of the walk from $(0, S_0)$ to (n, S_n) with r positive steps and s negative steps. If $S_n - S_0 = k$, then $r - s = k$ and $r + s = n$. Hence, $r = \frac{1}{2}(n + k)$ and $s = \frac{1}{2}(n - k)$. There are $\binom{n}{r}$ such paths, and each has the same probability, namely, $p^r q^s$. Hence,

$$\mathbf{P}(S_n - S_0 = k) = \binom{n}{r} p^r q^s,$$

which is (4).

Alternatively, we can simply observe that $\frac{1}{2}(S_n - S_0 + n)$ is a binomial random variable with parameters n and p, and (4) follows. ■

As suggested by the gambler's ruin problem, we are interested in the first passage times of random walks.

(5) Definition *Let $(S_n; n \geq 0)$ be a simple random walk with $S_0 = i$. Then, the* first passage time *from i to k is the random variable*

$$T_{ik} = \min\{n > 0; S_n = k\}. \quad \blacktriangle$$

When $i = k$, the random variable T_{kk} is called the *recurrence time* of k. We often denote T_{kk} by T_k. One obvious but important property of a first passage time T_{ik} is that steps after the first passage to k are independent of those before. It follows that we can write, for example,

$$T_{02} = T_{01} + T_{12}, \tag{6}$$

where T_{01} and T_{12} are independent. Furthermore, T_{12} and T_{01} have the same distribution because the X_i are identically distributed. These simple remarks are of great importance in examining the properties of the walk.

Our first result is striking.

(7) Theorem *If $p \geq q$, then T_{01} is certainly finite. If $p < q$, then T_{01} is finite with probability p/q.*

Proof Let $\mathbf{P}(T_{jk} < \infty) = r_{jk}$. Now let us condition on the first step of the walk from $S_0 = 0$, giving

$$r_{01} = \mathbf{P}(T_{01} < \infty) = p\mathbf{P}(T_{01} < \infty | X_1 = 1) + q\mathbf{P}(T_{01} < \infty | X_1 = -1). \tag{8}$$

On the one hand, if $X_1 = 1$, then $T_{01} = 1$. On the other hand, if $X_1 = -1$, then the walk has to go from -1 to 0 and then from 0 to $+1$ in a finite number of steps for $T_{01} < \infty$. Hence, as $r_{01} = r_{-10}$,

$$r_{01} = p + qr_{01}^2. \tag{9}$$

This has two roots $r_{01} = 1$ and $r_{01} = p/q$. It follows that if $p \geq q$, then the only root that is a probability is $r_{01} = 1$, as required.

If $p < q$, then it certainly seems plausible that $r_{01} = p/q$, but a little work is needed to prove it. This can be done in several ways. We choose to use the following interesting fact; that is, for all i,

$$\mathbf{E}\left(\left(\frac{q}{p}\right)^{X_i}\right) = p\left(\frac{q}{p}\right)^{+1} + q\left(\frac{q}{p}\right)^{-1} = 1. \tag{10}$$

Hence, if $S_0 = 0$,

$$\mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n}\right) = \prod_{i=1}^{n} \mathbf{E}\left(\left(\frac{q}{p}\right)^{X_i}\right) = 1. \tag{11}$$

Now denote T_{01} by T, and suppose that

$$p < q \quad \text{and} \quad r_{01} = 1. \tag{12}$$

Then conditioning on whether $T \leq n$ or $T > n$ gives

$$\begin{aligned} 1 &= \mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n}\right) \\ &= \mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n} \middle| T \leq n\right)\mathbf{P}(T \leq n) + \mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n} \middle| T > n\right)\mathbf{P}(T > n). \end{aligned} \tag{13}$$

Now if $T \leq n$, then $S_T = 1$ and $S_n = 1 + X_{T+1} + \cdots + X_n$. Hence,

$$\mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n} \middle| T \leq n\right) = \frac{q}{p}\mathbf{E}\left(\left(\frac{q}{p}\right)^{X_{T+1}+\cdots+X_n}\right) = \frac{q}{p} \quad \text{by (10)}.$$

Furthermore, if $T > n$, then $S_n \leq 0$, and so

$$\mathbf{E}\left(\left(\frac{q}{p}\right)^{S_n} \middle| T > n\right) \leq 1.$$

Hence, allowing $n \to \infty$ in (13) gives $1 = q/p + 0$. But this is impossible when $p < q$, so (12) must be impossible. Hence, when $p < q$, we must have $r_{01} = p/q$. ■

In the case when $p > q$, so that T_{01} is finite, it is natural to ask what is $\mathbf{E}(T_{01})$? If we knew that $\mathbf{E}(T_{01}) < \infty$, then we could write

$$\begin{aligned} \mathbf{E}(T_{01}) &= \mathbf{E}(\mathbf{E}(T_{01}|X_1)) = p\mathbf{E}(T_{01}|X_1 = 1) + q\mathbf{E}(T_{01}|X_1 = -1) \\ &= p + q(1 + \mathbf{E}(T_{-1,1})) = 1 + 2q\mathbf{E}(T_{01}) \quad \text{by (6)}. \end{aligned}$$

Hence,

$$\mathbf{E}(T_{01}) = \frac{1}{p - q}; \quad p > q, \tag{14}$$

as required.
It is not too difficult to show that $\mathbf{E}(T_{01}) < \infty$, as we now demonstrate.

(15) Theorem $\mathbf{E}(T_{01}) < \infty$ *when* $p > q$.

Proof

$$\begin{aligned} \mathbf{P}(T_{01} > n) &= \mathbf{P}(S_i \leq 0 \text{ for } 0 \leq i \leq n) \\ &\leq \mathbf{P}(S_n \leq 0) = \mathbf{P}\left(\frac{1}{2}(S_n + n) \leq \frac{n}{2}\right) \end{aligned}$$

$$= \mathbf{P}\left(\left(\frac{q}{p}\right)^{\frac{1}{2}(S_n+n)} \geq \left(\frac{q}{p}\right)^{n/2}\right) \text{ since } p > q,$$
$$\leq \mathbf{E}\left(\left(\frac{q}{p}\right)^{\frac{1}{2}(S_n+n)}\right)\left(\frac{q}{p}\right)^{-n/2}$$

by the basic inequality, Theorem 4.6.1.

Now we recall the observation in Theorem 4 that $\frac{1}{2}(S_n + n)$ has the $B(n, p)$ mass function. Hence,

$$\mathbf{E}\left(\left(\frac{q}{p}\right)^{\frac{1}{2}(S_n+n)}\right) = \left(q + p\,.\,\frac{q}{p}\right)^n = (2q)^n. \tag{16}$$

Therefore, finally,

$$\mathbf{E}(T_{01}) = \sum_{n=0}^{\infty} \mathbf{P}(T_{01} > n) \leq \sum_{n=0}^{\infty} (2q)^n \left(\frac{p}{q}\right)^{n/2} = \frac{1}{1 - 2(pq)^{\frac{1}{2}}} \quad \text{since } pq < \frac{1}{4}$$

This establishes (14), as required. ■

There are one or two gaps in the above; for example, we do not yet know either the mass function of T_{01} or $\mathbf{E}(T_{01})$ in the case $p = q = \frac{1}{2}$. Both of these can be filled by the following beautiful theorem.

(17) Hitting Time Theorem Let $(S_n; n \geq 0)$ be a simple random walk with $S_0 = 0$. Let T_{0b} be the first passage time from 0 to $b > 0$, with mass function $f_{0b}(n)$. Then

$$f_{0b}(n) = \mathbf{P}(T_{0b} = n) = \frac{b}{n}\mathbf{P}(S_n = b) = \frac{b}{n}\binom{n}{\frac{1}{2}(n+b)} p^{\frac{1}{2}(n+b)} q^{\frac{1}{2}(n-b)}.$$

The proof of (17) relies on the following lemma, which is of considerable interest in its own right. First, we observe that the number of paths of the walk from $(0, 0)$ to $(n - 1, b + 1)$ is denoted by $N_{n-1}(0, b + 1)$, and we have

$$N_{n-1}(0, b+1) = \binom{n-1}{\frac{1}{2}(n-b)-1}. \tag{18}$$

(19) Lemma: The Reflection Principle *Let $N^b_{n-1}(0, b - 1)$ be the number of paths from $(0, 0)$ to $(n - 1, b - 1)$ that pass through b at least once. Then*

$$N^b_{n-1}(0, b-1) = N_{n-1}(0, b+1). \tag{20}$$ ●

Proof Let π be a path that visits b on its journey from $(0, 0)$ to $(n - 1, b - 1)$. Let L be the occasion of its last visit. Now reflect that part of the walk after L in the line $y = b$. This yields a path π' from $(0, 0)$ to $(n - 1, b + 1)$. Conversely, for any path from $(0, 0)$ to $(n - 1, b + 1)$, we may reflect the segment in $y = b$ after its last visit

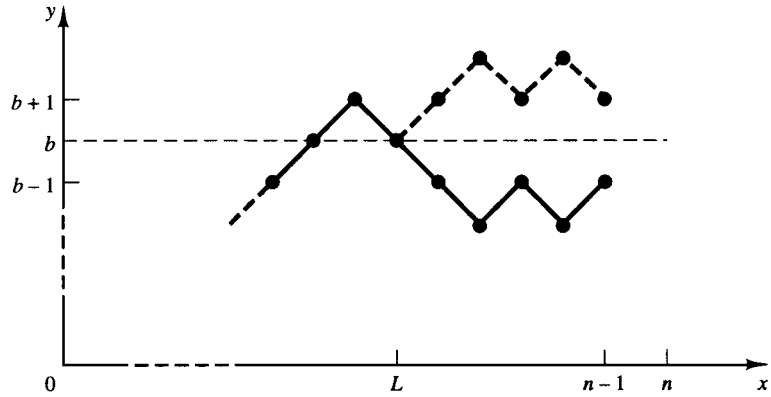


Figure 5.2 The solid line is the path of the walk: the dashed line is the reflection in $y = b$ of that part of the walk after its last visit to b before n, at time L.

to b, to give a path π from (0, 0) to $(n-1, b-1)$. These two sets are thus in one–one correspondence, and (20) follows. Figure 5.2 illustrates the reflection. ■

Proof of (17): Hitting Time Theorem If $T_{0b} = n$, then we must have $X_n = +1$ and $S_{n-1} = b - 1$. Now there are $N_{n-1}(0, b-1)$ paths from (0, 0) to $(n-1, b-1)$ of which $N^b_{n-1}(0, b-1)$ visit b on route. Each such path has probability $p^{\frac{1}{2}(n+b)-1}q^{\frac{1}{2}(n-b)}$. Hence, using the reflection principle,

$$\begin{aligned}
\mathbf{P}(T_{0b} = n) &= p(N_{n-1}(0, b-1) - N_{n-1}(0, b+1))p^{\frac{1}{2}(n+b)-1}q^{\frac{1}{2}(n-b)} \\
&= \left(\binom{n-1}{\frac{1}{2}(n+b)-1} - \binom{n-1}{\frac{1}{2}(n+b)}\right) p^{\frac{1}{2}(n+b)}q^{\frac{1}{2}(n-b)} \\
&= \frac{b}{n}\binom{n}{\frac{1}{2}(n+b)} p^{\frac{1}{2}(n+b)}q^{\frac{1}{2}(n-b)} \\
&= \frac{b}{n}\mathbf{P}(S_n = b), \quad \text{by (4).}
\end{aligned}$$

■

Because a similar argument works for negative values of b, we have

$$\mathbf{P}(T_{0b} = n) = \frac{|b|}{n}\mathbf{P}(S_n = b) \tag{21}$$

and $\mathbf{E}(T_{0b}) = \sum_{n=1}^{\infty} |b|\mathbf{P}(S_n = b)$.

(22) Example: Symmetric Random Walk When $p = q$, the simple random walk is said to be symmetric. In this case,

$$\begin{aligned}
\mathbf{E}(T_{01}) &= \sum_{m=0}^{\infty} \mathbf{P}(S_{2m+1} = 1) = \sum_{m=0}^{\infty} \binom{2m+1}{m+1} 2^{-(2m+1)} \\
&\geq \sum_{m=0}^{\infty} \frac{2m(2m-2)\ldots 2.1}{(m+1)(m)\ldots 1} 2^{-m} = \sum_{m=0}^{\infty} \frac{1}{m+1} = \infty.
\end{aligned}$$

Hence, the symmetric random walk has the interesting property that $\mathbf{P}(T_{01} < \infty) = 1$, but

$$\mathbf{E}(T_{01}) = \infty. \tag{23}$$

●

(24) Example: Conditioned Random Walk Now, of course, when $p < q$ we know that $\mathbf{P}(T_{01} < \infty) = p/q < 1$, so T_{01} has no expectation. But consider S_n conditional on the event that $T_{01} < \infty$. In this case, by conditional probability,

$$\mathbf{P}(X_1 = +1 | T_{01} < \infty) = \frac{\mathbf{P}(T_{01} < \infty | X_1 = 1)\mathbf{P}(X_1 = 1)}{\mathbf{P}(T_{01} < \infty)} = p / \left(\frac{p}{q}\right) = q.$$

Likewise, $\mathbf{P}(X_1 = -1 | T_{01} < \infty) = p$. Hence, if we knew that $\mathbf{E}(T_{01} | T_{01} < \infty)$ were finite, then by conditioning on the first step,

$$\begin{aligned} \mathbf{E}(T_{01} | T_{01} < \infty) &= q + p\mathbf{E}(T_{01} | X_1 = -1; T_{01} < \infty) \\ &= 1 + p2\mathbf{E}(T_{01} | T_{01} < \infty), \quad \text{by (6).} \end{aligned}$$

Hence, when $q > p$,

$$\mathbf{E}(T_{01} | T_{01} < \infty) = \frac{1}{q - p}. \tag{25}$$

It is straightforward to use Theorem 17 to establish that $\mathbf{E}(T_{01} | T_{01} < \infty) < \infty$ (an exercise for you) and (25) is proved. Together with (14) and Example 22, this shows that for any value of p and $b > 0$,

$$\mathbf{E}(T_{0b} | T_{0b} < \infty) = \frac{b}{|p - q|}. \tag{26}$$

●

We may also consider recurrence times.

(27) Example Let T_0 be the recurrence time of 0. Show that $\mathbf{P}(T_0 < \infty) = 1 - |p - q|$ and

$$\mathbf{E}(T_0 | T_0 < \infty) = 1 + \frac{1}{|p - q|}.$$

Solution Just consider T_0 conditional on the outcome of the first step, and then use what we know about T_{01} and T_{10}. You fill in the details. ●

Finally, we prove a famous result, the so-called

(28) Ballot Theorem Let $S_n = \sum_1^n X_i$ be a simple random walk with $S_0 = 0$. Then

$$\mathbf{P}\left(\prod_1^{2n-1} S_i \neq 0 | S_{2n} = 2r \right) = \frac{r}{n}.$$

Proof We count paths as we did in the hitting time theorem. What is the number $N^0_{2n-1}(1, 2r)$ of paths from $(1, 1)$ to $(2n, 2r)$ that visit the origin? We can reflect the

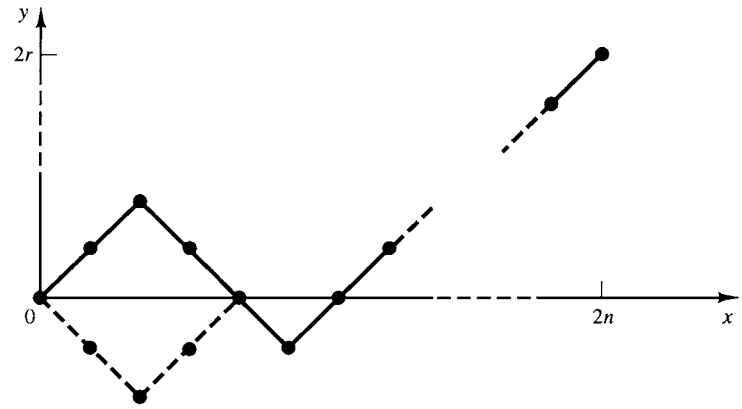


Figure 5.3 The ballot theorem The solid line is a path of the walk; the dashed line is the reflection in the x-axis of that part of the walk before its first visit to zero.

walk before its first zero in the x-axis, see Figure 5.3, and this shows that $N^0_{2n-1}(1, 2r) = N_{2n-1}(-1, 2r)$. Because all $N_{2n}(0, 2r)$ paths from $(0, 0)$ to $(2n, 2r)$ are equally likely, it follows that the required probability is

$$\frac{N_{2n-1}(1,2r) - N^0_{2n-1}(1,2r)}{N_{2n}(0,2r)} = \frac{N_{2n-1}(1,2r) - N_{2n-1}(-1,2r)}{N_{2n}(0,2r)}$$
$$= \frac{\binom{2n-1}{n+r-1} - \binom{2n-1}{n+r}}{\binom{2n}{n+r}} = \frac{2r}{2n}. \qquad \blacksquare$$

The following application explains the name of the theorem.

Example: Ballot In an election, candidate A secures a votes and candidate B secures b votes. What is the probability that A is ahead throughout the count? By the above argument, this probability is $(a - b)/(a + b)$ when $a > b$. ●

5.7 Martingales

In this section, we consider a remarkably useful class of random processes called martingales. They arise naturally as general models for fair games, but turn up in all kinds of unexpected places. In particular, they are used extensively in modern financial mathematics, but it is beyond our scope to explore this area in great detail. We begin with this:

(1) **Definition** *A collection* $(S_n; n \geq 0)$ *of random variables is a* martingale *if, for all* n,

(a) $\mathbf{E}|S_n| < \infty$.
(b) $\mathbf{E}(S_{n+1}|S_0, S_1, \ldots, S_n) = S_n$. ▲

6
Generating Functions and Their Applications

> Everything future is to be estimated by a wise man, in proportion to the probability of attaining it, and its value when attained.
>
> *Samuel Johnson,* [*The Rambler, 20*]

This chapter deals with a special subject and may be omitted on a first reading. Its contents are important and useful, but are not a prerequisite for most of the following chapters.

6.1 Introduction

In Chapter 3, we found that generating functions can provide elegant and concise methods for handling collections of real numbers. The mass function of an integer valued random variable is such a collection, and so we may anticipate (correctly as it turns out) that the following generating function will be useful.

(1) **Definition** *The* probability generating function *$G(s)$ of the integer valued random variable X is defined by*

$$G(s) = \sum_k \mathbf{P}(X = k)s^k. \qquad \blacktriangle$$

Because all random variables in this chapter are integer valued, this is not again mentioned explicitly.

(2) **Example** Let X be uniformly distributed in $\{-a, -a+1, \ldots, b-1, b\}$, where $a, b > 0$. Then provided $s \neq 1$,

$$G(s) = \sum_{k=-a}^{b} \frac{1}{a+b+1} s^k = \frac{s^{-a} - s^{b+1}}{(a+b+1)(1-s)}. \qquad \bullet$$

Notice that, by Theorem 4.3.4, we have from Definition 1 of $G(s)$ that

(3) $$G(s) = \mathbf{E}(s^X);$$

this is a particularly useful representation of $G(s)$, and we use it a great deal in what follows.

For example, suppose we seek the probability generating function of $Y = X + a$, where a is constant. Using (3), we can write

$$G_Y(s) = \mathbf{E}(s^Y) = \mathbf{E}(s^{X+a}) = s^a G_X(s).$$

We will see many other applications of (3) later. When X is defective (that is, when $\mathbf{P}(|X| < \infty) < 1$), the representation (3) can still be used, provided that we remember that the expectation is taken over the finite part of the distribution of X.

We write $G_X(s)$ when we want to stress the role of X; and for brevity, $G_X(s)$ is sometimes known as the p.g.f. of X.

Obviously, if $\mathbf{P}(|X| < \infty) = 1$, then

$$G_X(1) = \sum_k \mathbf{P}(X = k) = 1.$$

To sum up, if X is finite with probability 1, then $G_X(s)$ is a power series in s with nonnegative coefficients such that $G_X(1) = 1$.

Conversely, if $G(s)$ is a power series with nonnegative coefficients such that $G(1) = 1$, then G is the p.g.f. of some integer valued random variable X, which is finite with probability 1.

(4) Example Let $G(s) = (a + bs)/(1 - cs)$. When is G the p.g.f. of a finite integer valued random variable X?

Solution First, we note that if X is finite then $G(1) = 1$, and so $a + b + c = 1$. Now we need to consider various cases.

(i) If $0 \le c < 1$, then we can write, for any n,

$$\begin{aligned} G(s) &= (a + bs)(1 + cs + \cdots + (cs)^n) + \frac{a + bs}{1 - cs}(cs)^{n+1} \\ &= a + (b + ac)s + (b + ac)cs^2 + \cdots + (b + ac)c^{n-1}s^n + bc^n s^{n+1} + \frac{a + bs}{1 - cs}(cs)^{n+1}. \end{aligned}$$

For $|s| < c^{-1}$, we can let $n \to \infty$ to obtain a series expansion of $G(s)$. This has the required properties of a p.g.f. if $1 \ge a \ge 0$ and $1 \ge b + ac \ge 0$. In this case, X is a nonnegative random variable.

(ii) If $c = 1$, then $a = -b = 1$. In this case X is zero with probability 1.

(iii) If $c > 1$, then we can use a method similar to that of (i) to obtain a different series expansion of $G(s)$, that is:

$$G = -\frac{b}{c} - \frac{ac + b}{c^2 s} - \frac{ac + b}{c^3 s^2} - \cdots$$

This is a p.g.f. if $-c \le b \le 0$ and $-c^2 \le b + ac \le 0$. In this case, X is nonpositive.

(iv) If $c < 0$, then $a = 1$ and $b = c$. In this case, X is zero with probability 1.

See Example 14 for more insight into the nature of this probability generating function. ●

Another useful theorem is 3.6.7, which we restate here.

(5) Theorem *Let X be a random variable with mass function $f(k)$, and suppose that $a \le X \le b$. Let*

$$t_n = \mathbf{P}(X > n) = \sum_{k=n+1}^{b} f(k).$$

Define the tail generating function
$T(s) = \sum_{a}^{b-1} s^n t_n$. *Then, whenever both sides exist,*

$$(1-s)T(s) = s^a - G(s). \tag{6}$$

In particular, if $X \ge 0$, then

$$(1-s)T(s) = 1 - G(s). \tag{7}$$

Proof The left-hand side of (7) may be written as

$$\begin{aligned}(1-s)\sum_{n=a}^{b-1}\mathbf{P}(X>n)s^n &= \sum_{n=a}^{b-1} s^n\mathbf{P}(X>n) - \sum_{n=a}^{b-1} s^{n+1}\mathbf{P}(X>n)\\ &= \sum_{n=a+1}^{b-1} s^n(\mathbf{P}(X>n)-\mathbf{P}(X>n-1)) + s^a\mathbf{P}(X>a)\\ &\quad - s^b\mathbf{P}(X>b-1)\\ &= s^a - \sum_{a}^{b}\mathbf{P}(X=n)s^n = s^a - G(s),\end{aligned}$$

as required. ■

(8) Example 2 Revisited Here X is uniform on $\{-a, \dots, b\}$, and so

$$\begin{aligned}T_X(s) &= \frac{s^{-a}}{1-s} - \frac{(s^{-a}-s^{b-1})}{(a+b+1)(1-s)^2}\\ &= \frac{(1-s)(a+b)s^{-a} + s^{b+1} - s^{-a+1}}{(a+b+1)(1-s)^2}.\end{aligned}$$ ●

More generally, we can show that the identity (7) holds for unbounded nonnegative random variables. One way of doing this is to observe that the coefficients of s^n on each side are equal for all n, and then use a standard theorem about power series.

(9) Example Let X be geometric with mass function $f(k) = (1-q)q^{k-1}; k \ge 1, 0 < q < 1$. Then

$$G(s) = \sum_{k=1}^{\infty}(1-q)q^{k-1}s^k = \frac{(1-q)s}{1-qs},$$

and

$$(1-s)T(s) = 1 - \frac{(1-q)s}{1-qs} = \frac{1-s}{1-qs}.$$

Thus,

$$T(s) = \frac{1}{1-qs}. \quad \bullet$$

For future reference, we record the following trivial corollary of (7); that is, if $\mathbf{P}(0 \le X < \infty) = 1$, then

$$\sum_{j=0}^{\infty} s^j \mathbf{P}(X \le j) = \frac{G_X(s)}{1-s}. \tag{10}$$

It is useful to bear in mind that conditional probability mass functions also have generating functions. Thus, if A is some event, we can write $\mathbf{P}(X = k|A) = f(k|A)$ and define the generating function

$$G_{X|A}(s) = \sum_k f(k|A)s^k = \mathbf{E}(s^X|A),$$

in the usual notation.

If $(A_i; i \ge 1)$ is a collection of disjoint events with $\bigcup_i A_i = \Omega$, then it is easy to show that

$$\mathbf{E}(s^X) = \sum_i \mathbf{E}(s^X|A_i)\mathbf{P}(A_i). \tag{11}$$

This result is often useful in finding $\mathbf{E}(s^X)$.

If the random variables X and Y are jointly distributed, then in like manner we have

$$\mathbf{E}(s^X|Y = y) = \sum_k s^k \mathbf{P}(X = k|Y = y) = \sum_k s^k \frac{f(k, y)}{f_Y(y)}; \quad f_Y(y) > 0. \tag{12}$$

As y runs over all the possible values of Y, this yields the conditional p.g.f. of X given Y

$$G_{X|Y}(s) = \mathbf{E}(s^X|Y). \tag{13}$$

We therefore have the useful result:

$$G_X(s) = \mathbf{E}(G_{X|Y}(s)) = \mathbf{E}(\mathbf{E}(s^X|Y)).$$

(14) Example 4 Revisited Suppose we have two biased coins; the first shows a head with probability a, and the second shows a head with probability $1 - c$. The first coin is tossed and, if it shows a tail then the second coin is tossed repeatedly until a head is shown. Let X be the number of times the second coin is tossed. What is $G_X(s)$?

Solution Let H be the event that the first coin shows a head. If H occurs then $X = 0$, so $\mathbf{E}(s^X|H) = 1$. If H^c occurs, then X is geometric with $f_X(k) = (1-c)c^{k-1}$; $k \ge 1$. Hence, by Example 9,

$$\mathbf{E}(s^X|H^c) = \frac{(1-c)s}{1-cs}.$$

Therefore, by (11),

$$\mathbf{E}(s^X) = a + \frac{(1-a)(1-c)s}{1-cs} = \frac{a + (1-a-c)s}{1-cs}.$$

Looking back, we see that this is the generating function considered in Example 4, case (i). It follows that we can think of $(a + bs)/(1 - cs)$ as being the p.g.f. of a random variable X, which is either zero with probability a, or with probability $1 - a$ is a geometric random variable with parameter c. Such random variables arise quite naturally in applications. ●

(15) Example A biased coin is tossed repeatedly until the first occasion when r consecutive heads have resulted. Let X be the number of tosses required. Find $\mathbf{E}(s^X)$.

Solution We suppose that the chance of a head is p, and note that if the first i tosses are $i - 1$ heads followed by a tail, then the further number of tosses required has the same mass function as X. Hence, with an obvious notation:

$$\mathbf{E}(s^X | H^{i-1}T) = s^i \mathbf{E}(s^X); \quad 1 \le i \le r.$$

Also, $\mathbf{E}(s^X | H^r) = s^r$. It follows that

$$\mathbf{E}(s^X) = \sum_{i=1}^{r} qp^{i-1}s^i \mathbf{E}(s^X) + p^r s^r$$

and so

$$\mathbf{E}(s^X)\left(1 - qs\sum_{i=0}^{r-1}(ps)^i\right) = p^r s^r.$$

Hence,

$$\mathbf{E}(s^X) = \frac{p^r s^r (1-ps)}{1 - s + qp^r s^{r+1}}.$$

We discover different methods for proving this later. ●

6.2 Moments and the Probability Generating Function

For the remainder of this chapter, random variables are assumed to be nonnegative unless stated otherwise. In this case, whenever $|s| \le 1$,

$$|G_X(s)| = \left|\sum_0^\infty f(k)s^k\right| \le \sum_0^\infty f(k)|s^k| \le \sum_0^\infty f(k) = 1.$$

This simple property has enormous consequences for the p.g.f. $G(s)$. These are fully explored in textbooks on calculus and analysis, so we merely state the most important relevant results here. First, we state without proof:

(1) Theorem *The function $G(s)$ is differentiable for $|s| < 1$ and its derivative is*

$$G'(s) = \sum_{n=1}^{\infty} nf(n)s^{n-1} < \infty.$$

At $s = 1$,

$$G'(1) = \lim_{s \uparrow 1} \sum_{n=1}^{\infty} n f(n) s^{n-1} \tag{2}$$

whether the limit is finite or not. More generally, it follows that for $k \geq 1$,

$$G^{(k)}(s) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} f(n) s^{n-k}, \quad |s| < 1 \tag{3}$$

and

$$G^{(k)}(1) = \lim_{s \uparrow 1} \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} f(n) s^{n-k}. \tag{4}$$

Second, it follows that $G(s)$ *determines the collection* $(f(k); k \geq 0)$.

(5) Theorem (Uniqueness) *Let* X *and* Y *have generating functions* $G_X(s)$ *and* $G_Y(s)$. *If for some* $G(s)$, *we have*

$$G_X(s) = G_Y(s) = G(s) \text{ for } |s| < 1,$$

then X *and* Y *have the same mass function.*

Proof This follows from (3) because both $f_X(k)$ and $f_Y(k)$ are given by

$$f_X(k) = \frac{G^{(k)}(0)}{k!} = f_Y(k)$$

for all k. ■

Third, it follows that we can obtain all the moments of X from $G(s)$.

(6) Theorem *If* X *has p.g.f.* $G(s)$, *then*

$$\mathbf{E}(X) = G'(1); \tag{7}$$

more generally, the k*th* factorial moment *is*

$$\mu^{(k)} = \mathbf{E}(X(X-1)\dots(X-k+1)) = G^{(k)}(1); \tag{8}$$

and, in particular,

$$\operatorname{var}(X) = G''(1) + G'(1) - (G'(1))^2. \tag{9}$$

Proof Equation (7) is a trivial consequence of (2), and (8) follows from (4). To see (9), write

$$\begin{aligned} \operatorname{var}(X) &= \mathbf{E}(X - \mathbf{E}(X))^2 = \mathbf{E}(X^2) - (G'(1))^2 \\ &= \mathbf{E}(X(X-1) + X) - (G'(1))^2, \end{aligned} \tag{10}$$

as required. ■

Just as (10) gives the second moment and second central moment in terms of the first two factorial moments; likewise, σ_k and μ_k may be obtained in principle in terms of $(\mu^{(k)}; k \geq 1)$.

(11) Example: Binomial p.g.f. Let X have a binomial distribution with parameters n and p. Then, with $q = 1 - p$ as usual,

$$G(s) = \sum_0^n \binom{n}{k} q^{n-k} p^k s^k = (q + ps)^n.$$

Now using (8), we have

$$\mu^{(k)} = \begin{cases} \dfrac{n!}{(n-k)!} p^k & 1 \leq k \leq n, \\ 0 & k > n. \end{cases}$$

Hence, by (9), $\text{var}\,(X) = n(n-1)p^2 + np - (np)^2) = npq$. ●

(12) Example: Poisson p.g.f. Let X have a Poisson distribution with parameter λ. Then

$$G(s) = \sum_0^\infty e^{-\lambda} \frac{\lambda^k}{k!} s^k = e^{+\lambda(s-1)}.$$

Hence, we find that $\mu^{(k)} = \lambda^k$, for $k \geq 1$. ●

Moments can also be obtained from the tail generating function $T(s)$ defined in Theorem 6.1.5.

(13) Theorem *Let X be a random variable with*

$$T(s) = \sum_0^\infty s^k \mathbf{P}(X > k).$$

Then

$$\mathbf{E}(X) = T(1) \tag{14}$$

and, if $\mathbf{E}(X) < \infty$,

$$\text{var}\,(X) = 2T'(1) + T(1) - T(1)^2. \tag{15}$$

Proof By L'Hôpital's rule,

$$T(1) = \lim_{s \uparrow 1} \frac{1 - G(s)}{1 - s} = G'(1) = \mathbf{E}(X), \quad \text{by (7)}.$$

Likewise, differentiating (6.1.7) yields

$$T'(1) = \lim_{s \uparrow 1} \left(\frac{1 - G(s)}{(1-s)^2} - \frac{G'(s)}{1-s} \right) = \frac{G''(1)}{2}, \quad \text{by L'Hôpital's rule,}$$

and the result follows using Theorem 6. ■

More generally, a straightforward extension of this theorem shows that

$$\mu^{(k)} = kT^{(k-1)}(1). \tag{16}$$

(17) Example: Geometric p.g.f. Let X have a geometric distribution with mass function $f(k) = (1-q)q^{k-1}$, $k \geq 1; 0 < q < 1$. Then, by Example 6.1.9,

$$T(s) = \frac{1}{1-qs}.$$

Hence, by Theorem 13,

$$\mathbf{E}(X) = T(1) = \frac{1}{1-q},$$

and likewise

$$\text{var}\,(X) = 2T'(1) + T(1) - T(1)^2 = \frac{q}{(1-q)^2}.$$

From (16),

$$\mu^{(k)} = \frac{kq^{k-1}}{(1-q)^k}.$$ ●

We conclude this section with a note about defective probability mass functions. If X is a nonnegative random variable such that $\sum_{k=0}^{\infty} f(k) < 1$, then it still makes sense to define the generating function $G(s) = \sum_{k=0}^{\infty} s^k f(k)$. Furthermore, if $\sum_k kf(k) < \infty$, then $G'(1) = \sum_k kf(k)$. However, this is not now the expectation $\mathbf{E}(X)$, but rather the "defective" expectation

$$\mathbf{E}(XI\{X < \infty\}) = \mathbf{E}(X; X < \infty) = \mathbf{E}(X|X < \infty)\mathbf{P}(X < \infty),$$

where $I\{X < \infty\}$ is the indicator of the event that X is finite.

In the general case, we have likewise

$$G'(1) = \mathbf{E}(X||X| < \infty)\mathbf{P}(|X| < \infty)$$

when the expectation exists. In such cases, it can be of interest to calculate

$$\mathbf{E}(X|X < \infty) = \frac{G'(1)}{G(1)}.$$

6.3 Sums of Independent Random Variables

If X and Y are independent, then the mass function of their sum $Z = X + Y$ is

$$f_Z(k) = \sum_j f_X(j) f_Y(k-j). \tag{1}$$

Practical folk (such as statisticians and the like) are frequently interested in the sum of n independent random variables:

$$S = \sum_1^n X_i. \tag{2}$$

The prospect of performing the summation in (1) on $n-1$ occasions to find $f_S(k)$ is not an attractive one. The next theorem renders it unnecessary in many important cases.

(3) Theorem *(a) Let X_1 and X_2 be independent with generating functions $G_1(s)$ and $G_2(s)$, respectively. Then the sum $Z = X_1 + X_2$ has generating function*

(4) $$G(s) = G_1(s)G_2(s).$$

(b) More generally, if $(X_i; 1 \le i \le n)$ are independent with generating functions $(G_i(s); 1 \le i \le n)$, then the sum $Z = \sum_1^n X_i$ has generating function

$$G_Z(s) = \prod_{i=1}^{n} G_i(s).$$

Proof (a) Because X_1 and X_2 are independent, s^{X_1} and s^{X_2} are also independent. Hence,

$$\begin{aligned} G_Z(s) &= \mathbf{E}(s^{X_1+X_2}) = \mathbf{E}(s^{X_1})\mathbf{E}(s^{X_2}) \quad \text{by Theorem 5.3.8} \\ &= G_1(s)G_2(s). \end{aligned}$$

Part (b) is proved similarly. ■

Example: Binomial Sum Let X and Y be independent and binomially distributed with parameters (m, p) and (n, p), respectively. Then recalling Example 6.2.11, we have

$$\begin{aligned} G_{X+Y}(s) &= \mathbf{E}(s^{X+Y}) = \mathbf{E}(s^X)\mathbf{E}(s^Y) \quad \text{by independence} \\ &= (1 - p + ps)^{m+n}. \end{aligned}$$

Hence, $X + Y$ is binomially distributed with parameters $m + n$ and p, using Theorem 6.2.5, the uniqueness theorem. ●

(5) Example Let $(X_i; i \ge 1)$ be independent Poisson random variables having respective parameters $(\lambda_i; i \ge 1)$. Find the mass function of $Z = \sum_{i=1}^n X_i$.

Solution Reproducing the argument of the above theorem, we have:

$$\begin{aligned} G_Z(s) = \mathbf{E}(s^Z) &= \prod_{i=1}^{n} \mathbf{E}(s^{X_i}) \qquad \text{by independence} \\ &= \exp\left(\sum_{i=1}^{n} \lambda_i(s-1)\right) \quad \text{by Example 6.2.12.} \end{aligned}$$

Thus, Z is Poisson with parameter $\sum_{i=1}^n \lambda_i$, by the uniqueness theorem. ●

(6) Example Let $(X_i; i \geq 1)$ be independently and uniformly distributed on $\{1, 2, \ldots, n\}$. Let $S_k = \sum_{i=1}^{k} X_i$, and define

$$T_n = \min\{k: S_k > n\}.$$

(Thus, T_n is the smallest number of the X_i required to achieve a sum exceeding n.) Find the mass function and p.g.f. of T_n, and hence calculate $\mathbf{E}(T_n)$ and $\text{var}(T_n)$.

Solution First, we observe that $T_n \geq j + 1$ if and only if $S_j \leq n$. Therefore,

$$\mathbf{P}(T_n \geq j + 1) = \mathbf{P}(S_j \leq n). \tag{7}$$

Now, by independence,

$$\mathbf{E}(z^{S_j}) = (\mathbf{E}(z^{X_1}))^j = \frac{1}{n^j}\left(\frac{z - z^{n+1}}{1 - z}\right)^j \quad \text{by Example 6.1.2.}$$

Hence, by Example 6.1.9,

$$\sum_{k=j}^{\infty} z^k \mathbf{P}(S_j \leq k) = \left(\frac{z}{n}\right)^j \frac{(1 - z^n)^j}{(1 - z)^{j+1}}. \tag{8}$$

Equating coefficients of z^n on each side of (8) gives

$$\mathbf{P}(S_j \leq n) = \frac{1}{n^j}\binom{n}{j} = \mathbf{P}(T_n \geq j + 1) \quad \text{by (7).} \tag{9}$$

Hence,

$$\mathbf{P}(T_n = j) = \frac{1}{n^{j-1}}\binom{n}{j - 1} - \frac{1}{n^j}\binom{n}{j}.$$

From (9), T_n has tail generating function

$$\sum_{j=0}^{n} z^j \mathbf{P}(T_n > j) = \sum_{j=0}^{n} \left(\frac{z}{n}\right)^j \binom{n}{j} = \left(1 + \frac{z}{n}\right)^n.$$

Hence, from Theorem 6.2.13,

$$\mathbf{E}(T_n) = \left(1 + \frac{1}{n}\right)^n \tag{10}$$

and

$$\text{var}(T_n) = 2\left(1 + \frac{1}{n}\right)^{n-1} + \left(1 + \frac{1}{n}\right)^n - \left(1 + \frac{1}{n}\right)^{2n}. \tag{11}$$

Finally, T_n has p.g.f.

$$G(z) = 1 + (z - 1)\left(1 + \frac{z}{n}\right)^n. \tag{12}$$

●

Generating functions become even more useful when you are required to consider the sum of a random number of random variables.

(13) Theorem: Random Sum *Let N and $(X_i; i \geq 1)$ be independent random variables, and suppose that N is nonnegative and that for all i*

(14) $$\mathbf{E}(s^{X_i}) = G(s).$$

Then the sum $Z = \sum_{i=1}^{N} X_i$ has generating function

$$G_Z(s) = G_N(G(s)).$$

Proof By conditional expectation,

$$\begin{aligned} \mathbf{E}(s^Z) &= \mathbf{E}(\mathbf{E}(s^Z|N)) = \mathbf{E}(\mathbf{E}(s^{X_1}) \dots \mathbf{E}(S^{X_N})) && \text{by independence} \\ &= \mathbf{E}(G(s)^N) && \text{by (14)} \\ &= G_N(G(s)) \end{aligned}$$

by Definition 6.1.3, and the result follows. ■

(15) Example You toss a fair coin repeatedly. Each time it shows a tail you roll a fair die, when the coin first shows a head you stop. What is the p.g.f. of the total sum of the scores shown by the rolls of the die?

Solution As you know by now, the number N of tails shown has mass function $f_N(k) = (\frac{1}{2})^{k+1}; k \geq 0$, with generating function

$$G_N(s) = \frac{1}{2 - s}.$$

The score shown by each die has p.g.f.

$$G_X(s) = \frac{1}{6}\frac{s(1 - s^6)}{1 - s},$$

and so the p.g.f. of the total is given by Theorem 13 as

$$G(s) = \left(2 - \frac{1}{6}\frac{s(1 - s^6)}{1 - s}\right)^{-1}.$$ ●

(16) Example Let $Z = \sum_{i=1}^{N} X_i$, where

$$f_X(k) = \frac{-k^{-1}p^k}{\log(1 - p)}; \quad k \geq 1, 0 < p < 1$$

and $f_N(k) = \lambda^k e^{-\lambda}/k!; k \geq 1, 0 < \lambda$. Show that Z has a negative binomial mass function.

Solution It is easy to show that

$$G_X(s) = \frac{\log(1 - sp)}{\log(1 - p)}; \quad G_N(s) = e^{\lambda(s-1)}.$$

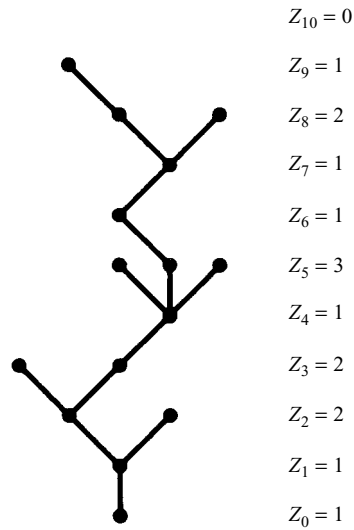


Figure 6.1 A realization of a branching process. The orientation of the diagram explains the name.

Hence,

$$G_Z(s) = e^{-\lambda} \exp(\lambda G_X(s)) = \left(\frac{1-p}{1-ps} \right)^{-\lambda(\log(1-p))^{-1}},$$

which is the p.g.f. of a negative binomial mass function. ●

(17) Example: Branching A collection of particles behaves in the following way. At time $n = 0$, there is one particle. At time $n = 1$, it is replaced by a random number X of particles, where X has mass function $f(k), k \geq 0$. At every subsequent time $n = 2, 3, \ldots$, each particle in existence at that time is replaced by a random number of new particles, called its *family*. All family sizes are independent, and they all have the same mass function as the first family X. [An example is given in Figure 6.1.]

Let the number of particles in existence at time n be Z_n. Find $\mathbf{E}(s^{Z_n})$ and $\lim_{n\to\infty} \mathbf{P}(Z_n = 0)$.

Solution Let $G(s) = \mathbf{E}(s^X)$ and $G_n(s) = \mathbf{E}(s^{Z_n})$. Let the family sizes of the particles existing at time n be $(X_j; 0 \leq j \leq Z_n)$. Then we obtain the attractive and useful representation

$$Z_{n+1} = \sum_{j=0}^{Z_n} X_j$$

and, by Theorem 13,

$$G_{n+1}(s) = G_n(G(s)).$$

(This basic argument is used repeatedly in the theory of branching processes.) Hence, $G_{n+1}(s)$ is the $(n+1)$th iterate of $G(.)$, that is to say:

$$G_{n+1}(s) = G(G(\ldots G(s)\ldots)), \quad n \geq 0. \tag{18}$$

Now let $\mathbf{P}(Z_n = 0) = \eta_n$, and define η to be the smallest nonnegative root of the equation

$$G(s) = s \tag{19}$$

We now show that

$$\lim_{n\to\infty} \eta_n = \eta. \tag{20}$$

First, we consider three trivial cases:

(i) If $f(0) = 0$ then $\eta_n = G_n(0) = 0 = \eta$.
(ii) If $f(0) = 1$ then $\eta_n = G_n(0) = 1 = \eta$.
(iii) If $f(0) + f(1) = 1$, with $f(0)f(1) \neq 0$, then

$$\eta_n = G_n(0) = 1 - (f(1))^n \to 1 = \eta.$$

Thus, (20) is true in each case. In what follows, we exclude these cases by requiring that $0 < f(0) < f(0) + f(1) < 1$.

Now note that $\{Z_n = 0\} \subseteq \{Z_{n+1} = 0\}$ and so, by Example 1.4.11,

$$\eta_n \leq \eta_{n+1} \leq 1.$$

Hence, $\lim_{n\to\infty} \eta_n$ exists; let us denote it by λ. By (18), $G_{n+1}(0) = G(G_n(0))$; now letting $n \to \infty$ and using the continuity of $G(s)$, we find that λ is a root of (19):

$$\lambda = G(\lambda).$$

However, if for some n, $\eta_n < \eta$ then, because $G(s)$ is increasing,

$$\eta_{n+1} = G(\eta_n) \leq G(\eta) = \eta.$$

But $\eta_0 = G(0) \leq G(\eta) = \eta$, and so $\eta_n < \eta$ for all n. Hence, $\lambda \leq \eta$ and so $\lambda = \eta$. ●

Once again, we conclude with a note about defective random variables. If X and Y are defective, then they are independent if $\mathbf{P}(X = i, Y = j) = \mathbf{P}(X = i)\mathbf{P}(Y = j)$ for all finite X and Y. Hence, we can still write

$$G_{X+Y}(s) = G_X(s)G_Y(s),$$

and we can denote this by

$$\mathbf{E}(s^{X+Y}) = \mathbf{E}(s^X)\mathbf{E}(s^Y).$$

if we remember that the expectation is taken only over the finite part of the distribution.

6.4 Moment Generating Functions

The moments $(\mu_k; k \geq 1)$ of a random variable X also form a collection of real numbers, so we may also expect their generating functions to be useful. In this case, it is convenient to use the exponential generating function of the collection $(\mu_k; k \geq 1)$.

(1) Definition *Let the random variable X have finite moments $\mu_k = \mathbf{E}(X^k)$ for all $k \geq 1$. Then the* moment generating function *(or m.g.f.) of X is the function $M_X(t)$ given by*

(2) $$M_X(t) = \sum_{k=0}^{\infty} \frac{\mu_k t^k}{k!}.$$ ▲

If X takes only a finite number of values, then we easily obtain the convenient representation

(3) $$M_X(t) = \sum_{k=0}^{\infty} \mathbf{E}\frac{(X^k)t^k}{k!} = \mathbf{E}\left(\sum_{k=0}^{\infty} \frac{(Xt)^k}{k!}\right) = \mathbf{E}(e^{Xt}).$$

More generally, (3) holds provided the moments μ_k do not get too large as k increases. For example, if $\sum_{k=0}^{\infty} |\mu_k|/k! < \infty$, then $M_X(t)$ exists for $|t| < 1$, and we can use the equivalent of Theorem 6.2.1. This yields

(4) $$\mu_k = M_X^{(k)}(0).$$

From (3), we find that

(5) $$M_X(t) = G_X(e^t),$$

where $G_X(s)$ is the p.g.f. of X.

(6) Example Let X have a negative binomial distribution with mass function $f(k) = \binom{n+k-1}{k} q^k p^n$, $k \geq 0$. By the negative binomial expansion,

$$G(s) = \sum_{k=0}^{\infty} p^n \binom{n+k-1}{k} q^k s^k = \left(\frac{p}{1-qs}\right)^n, \quad |s| < q^{-1}.$$

Then X has moment generating function

$$M(t) = \left(\frac{p}{1-qe^t}\right)^n, \quad t < -\log q.$$ ●

Let us consider an example in which X may take negative integer values.

(7) Example Let X have mass function

$$f(k) = \frac{1}{2} q^{|k|-1}(1-q); \quad k = \pm 1, \pm 2, \ldots,$$

where $0 < q < 1$. Then X has p.g.f.

$$\begin{aligned} G(s) &= \sum_{k\neq 0} s^k \frac{1}{2} q^{|k|-1}(1-q) \\ &= \frac{1}{2}(1-q)\left(\frac{s}{1-qs} + \frac{1}{s-q}\right), \quad q < |s| < q^{-1}. \end{aligned}$$

Hence, X has m.g.f.

$$M(t) = \frac{1}{2}(1-q)\left(\frac{e^t}{1-qe^t} + \frac{1}{e^t - q}\right), \quad \log q < t < -\log q.$$

In this case, $M(t)$ exists in an interval including the origin, and (4) holds. ●

The factorial moments $(\mu^{(k)}; k \geq 1)$ also have a generating function, which is related to the probability generating function as follows:

$$\begin{aligned} \sum_k \frac{\mu^{(k)} t^k}{k!} &= \sum_k \mathbf{E}\frac{(X(X-1)\ldots(X-k+1))t^k}{k!} \\ &= \sum_k \sum_{n=k}^{\infty} \frac{n!}{(n-k)!}\frac{\mathbf{P}(X=n)}{k!} t^k \\ &= \sum_{n=0}^{\infty} \mathbf{P}(X=n) \sum_{k=0}^{n} \binom{n}{k} t^k \\ &= \sum_n \mathbf{P}(X=n)(1+t)^n = G_X(1+t). \end{aligned} \tag{8}$$

The change in the order of summation is justified because the terms in the sum are all nonnegative.

Now let us make the important observation that both of these moment generating functions are useful for dealing with sums of independent random variables for essentially the same reasons that made the p.g.f. so useful. To see this, let X and Y be independent, and set $Z = X + Y$. Then

$$M_Z(t) = \mathbf{E}(e^{t(X+Y)}) = M_X(t)M_Y(t) \quad \text{by independence.} \tag{9}$$

Likewise,

$$G_Z(1+t) = \mathbf{E}((1+t)^Z) = G_X(1+t)G_Y(1+t). \tag{10}$$

Finally, we record the existence of yet another function that generates the moments of X, albeit indirectly.

(11) Definition *If the function*

$$\kappa(t) = \log(\mathbf{E}(e^{Xt})) = \log(M_X(t))$$

can be expanded in powers of t, in the form

$$\kappa(t) = \sum_{r=1}^{\infty} \kappa_r t^r / r!, \tag{12}$$

then it is called the generating function of the cumulants *$(\kappa_r; r \geq 1)$.* ▲

(13) Example If X is Poisson with parameter λ, then

$$\log(M_X(t)) = \log(\exp[\lambda(e^t - 1)]) = \lambda(e^t - 1) = \sum_{r=1}^{\infty} \frac{\lambda}{r!} t^r.$$

Hence, for all r, $\kappa_r = \lambda$. ●

6.5 Joint Generating Functions

Generating functions can be equally useful when we want to consider the joint behaviour of a number of random variables. Not surprisingly, we need a joint generating function.

(1) Definition A random vector $(X_1, \ldots, X_n)$, with joint mass function $f(x_1, \ldots, x_n)$, has a *joint probability generating function*

$$G_X(s) = G_{X_1, \ldots, X_n}(s_1, \ldots, s_n) = \sum_{x_1, x_2, \ldots, x_n} s^{x_1} s^{x_2} \ldots s^{x_n} f(x_1, \ldots, x_n). \quad ▲$$

By Theorem 5.3.1, we obtain the following useful representation of G,

$$G_X(s) = \mathbf{E}\left(s_1^{X_1} s_2^{X_2} \ldots s_n^{X_n}\right) = \mathbf{E}\left(\prod_{i=1}^{n} s_i^{X_i}\right).$$

(2) Example A coin shows heads with probability p or tails with probability $q = 1 - p$. If it is tossed n times, then the joint p.g.f. of the number X of heads and the number of tails is

$$\begin{aligned} G(s, t) = \mathbf{E}(s^X t^Y) &= t^n \mathbf{E}\left(\left(\frac{s}{t}\right)^X\right) = t^n \left(q + p\frac{s}{t}\right)^n \quad \text{because } X \text{ is binomial,} \\ &= (qt + ps)^n. \end{aligned}$$ ●

The fact that G is the nth power of $(qt + ps)$ suggests that independence could have been used to get this result. We use this idea in the next example.

(3) Example: de Moivre trials Each of a sequence of n independent trials results in a win, loss, or draw, with probabilities α, β, and γ respectively. Find the joint p.g.f. of the wins, losses, and draws, the so-called *trinomial* p.g.f.

Solution Let W_i, L_i, and D_i be the respective indicators on the ith trial of a win, loss, or draw. Then

$$\mathbf{E}(x^{W_i} y^{L_i} z^{D_i}) = \alpha x + \beta y + \gamma z.$$

But the required joint p.g.f. is

$$G(x, y, z) = \mathbf{E}\left(x^{\sum_1^n W_i} y^{\sum_1^n L_i} z^{\sum_1^n D_i}\right) = [\mathbf{E}(x^{W_i} y^{L_i} z^{D_i})]^n \quad \text{by independence,}$$
$$= (\alpha x + \beta y + \gamma z)^n. \qquad \bullet$$

Knowledge of the joint p.g.f. entails knowledge of all the separate p.g.f.s because, for example, if X and Y have joint p.g.f. $G(s, t)$, then

$$G_X(s) = \mathbf{E}(s^X) = \mathbf{E}(s^X 1^Y) = G(s, 1). \tag{4}$$

Likewise,

$$G_Y(t) = G(1, t). \tag{5}$$

Indeed, we can quickly obtain the p.g.f. of any linear combination of X and Y; for example, let $Z = aX + bY$, then

$$\mathbf{E}(s^Z) = \mathbf{E}(s^{aX+bY}) = \mathbf{E}(s^{aX} s^{bY}) = G(s^a, s^b). \tag{6}$$

Further, the joint generating function also provides us with the joint moments when they exist, in the same way as $G_X(s)$ provides the moments of X.

(7) Example Let X and Y have joint p.g.f. $G(s, t)$ and suppose that X and Y have finite variance. Then $\mathbf{E}(XY)$ exists (by the Cauchy–Schwarz inequality) and

$$\frac{\partial^2 G}{\partial s \partial t} = \frac{\partial^2}{\partial s \partial t}\mathbf{E}(s^X s^Y) = \mathbf{E}(XY s^{X-1} t^{Y-1}).$$

Hence,

$$\mathbf{E}(XY) = \left.\frac{\partial^2 G}{\partial s \partial t}\right|_{s=t=1}. \tag{8}$$

Likewise,

$$\mathbf{E}(X) = \left.\frac{\partial G}{\partial s}\right|_{s=t=1}, \quad \text{and} \quad \mathbf{E}(Y) = \left.\frac{\partial G}{\partial t}\right|_{s=t=1}.$$

Quite often, we write $G_{st}(s, t)$ for $\partial^2 G/\partial s \partial t$, and so on; in this form, the covariance of X and Y is given by

$$\operatorname{cov}(X, Y) = G_{st}(1, 1) - G_s(1, 1) G_t(1, 1). \tag{9}$$ $\bullet$

(10) Example 5.11 Revisited: Golf Recall that you play n holes of golf, each of which you independently win, lose, or tie, with probabilities p, q, and r, respectively. The numbers of wins, losses, and ties are X, Y, and Z, respectively, with $X + Y + Z = n$.

(a) Find $\rho(X, Y)$. (b) Find $\operatorname{var}(X - Y)$.

Solution (a) By Example 3 above, we calculate

$$\mathbf{E}(x^X y^Y z^Z) = (px + qy + rz)^n = G(x, y, z) \quad \text{say.}$$

Hence,

$$\mathbf{E}(X) = G_x(1, 1, 1) = np,$$

and

$$\text{var}\,(X) = G_{xx}(1, 1, 1) + G_x(1) - (G_x(1))^2 = np(1 - p),$$

and

$$\mathbf{E}(XY) = G_{xy}(1, 1, 1) = n(n - 1)pq.$$

Therefore, the correlation between X and Y is

(11)
$$\begin{aligned}\rho(X, Y) &= \frac{\text{cov}\,(X, Y)}{(\text{var}\,(X)\text{var}\,(Y))^{\frac{1}{2}}} \\ &= \frac{n(n-1)pq - n^2pq}{(n^2p(1-p)q(1-q))^{\frac{1}{2}}} = -\left(\frac{pq}{(1-p)(1-q)}\right)^{\frac{1}{2}}.\end{aligned}$$

You should compare the labour in this calculation with the more primitive techniques of Example 5.11.

(b) Using (6) with $a = 1, b = -1$, we have, on setting $W = X - Y$,

$$G_W(s) = \mathbf{E}(s^{X-Y}) = G(s, s^{-1}, 1) = (ps + qs^{-1} + r)^n.$$

Hence, $dG_W/ds = n(p - qs^{-2})(ps + qs^{-1} + r)^{n-1}$, and $\frac{d^2G_W}{ds^2} = n(n - 1)(p - qs^{-2})^2 (ps + qs^{-1} + r)^{n-2} + 2nqs^{-3}(ps + qs^{-1} + r)^{n-1}$.
Therefore,

$$\begin{aligned}\text{var}\,(W) &= n(n-1)(p-q)^2 + 2nq + n(p-q) - n^2(p-q)^2 \\ &= n(p + q - (p-q)^2).\end{aligned}$$

●

Finally, we record that joint generating functions provide a useful characterization of independence.

(12) **Theorem** *Let X and Y have joint p.g.f. $G(s, t)$. Then X and Y are independent if and only if*

(13)
$$G(s, t) = G(s, 1)G(1, t).$$

Proof If (13) holds, then equating coefficients of $s^j t^k$ gives

$$\mathbf{P}(X = j, Y = k) = \mathbf{P}(X = j)\mathbf{P}(Y = k),$$

as required. The converse is immediate by Theorem 5.3.8. ■

(14) **Example 5.5.8 Revisited: Eggs** Recall that the number X of eggs is Poisson with parameter λ, and eggs hatch independently with probability p. Let Y be the number that do hatch, and Z the number that do not. Show that Y and Z are independent, and also that $\rho(X, Y) = \sqrt{p}$.

Solution Conditional on $X = x$, the number Y of hatchings is binomial with p.g.f.

$$\mathbf{E}(s^Y|X = x) = (ps + 1 - p)^x. \tag{15}$$

Hence, by conditional expectation,

$$\begin{aligned}\mathbf{E}(y^Y z^Z) = \mathbf{E}(y^Y z^{X-Y}) &= \mathbf{E}\left(z^X \mathbf{E}\left(\left(\frac{y}{z}\right)^Y |X\right)\right)\\ &= \mathbf{E}\left(z^X\left(\frac{py}{z} + 1 - p\right)^X\right) \qquad \text{by (15)}\\ &= \exp\left(\lambda(py + (1-p)z - 1)\right) \quad \text{since } X \text{ is Poisson,}\\ &= e^{\lambda p(y-1)} e^{\lambda(1-p)(z-1)}.\end{aligned}$$

Hence, Y and Z are independent by Theorem 12. Furthermore, we see immediately that Y is Poisson with parameter λp.

To find $\rho(X, Y)$, we first find the joint p.g.f. of X and Y, again using conditional expectation. Thus,

$$\mathbf{E}(s^X y^Y) = \mathbf{E}(s^X \mathbf{E}(y^Y|X)) = \mathbf{E}(s^X(py + 1 - p)^X) = \exp\left(\lambda(s(py + 1 - p) - 1)\right).$$

Hence, using (7), $\mathbf{E}(XY) = \lambda^2 p + \lambda p$, and so, using the first part, $\rho(X, Y) = \sqrt{p}$. You should compare this with the method of Example 5.5.8. ●

(16) Example: Pairwise Independence Independent random variables X and Y each take the values $+1$ or -1 only, and $\mathbf{P}(X = 1) = a$, with $\mathbf{P}(Y = 1) = b$. Let $Z = XY$. Show that there are values of a and b such that X, Y, and Z are pairwise independent.

(17) Solution Consider the joint probability generating function of X and Z.

$$\begin{aligned}G(s, t) = \mathbf{E}(s^X t^Z) = \mathbf{E}(s^X t^{XY}) &= \mathbf{E}(\mathbf{E}(s^X t^{XY}|X))\\ &= \mathbf{E}\left(bs^X t^X + (1-b)\frac{s^X}{t^X}\right)\\ &= a\left(bst + (1-b)\frac{s}{t}\right) + (1-a)\left(\frac{b}{st} + (1-b)\frac{t}{s}\right)\\ &= \frac{abs^2t^2 + a(1-b)s^2 + (1-a)(1-b)t^2 + b(1-a)}{st}\\ &= \frac{1}{st}\left(as^2(bt^2 + 1 - b) + \frac{(1-a)(1-b)}{b}(bt^2 + 1 - b)\right.\\ &\quad \left. + b(1-a) - (1-b)^2(1-a)b^{-1}\right),\end{aligned}$$

which factorizes into a product of a function of s and a function of t if

$$b^2 - (1-b)^2 = 0,$$

that is if $b = \frac{1}{2}$. In this case, X and Z are independent. If $a = \frac{1}{2}$ then Y and Z are independent, and $a = b = \frac{1}{2}$ entails the pairwise independence of X, Y, and Z. ●

6.6 Sequences

In Section 4.5, we defined the convergence of a sequence of mass functions. This can be usefully connected to the convergence of corresponding sequences of generating functions. For sequences of probability generating functions, we have the following result, which we give without proof.

(1) Theorem *Let $f(k)$ be a probability mass function with generating function*

$$G(s) = \sum_0^\infty s^k f(k),$$

and suppose that for each $n \geq 1$, $f_n(k)$ is a probability mass function with generating function

$$G_n(s) = \sum_0^\infty s^k f_n(k).$$

Then, as $n \to \infty$, $f_n(k) \to f(k)$ for all k, if and only if $G_n(s) \to G(s)$ for all $0 < s < 1$.

We now use this to prove a result, which we have already shown by more primitive methods.

(2) Example Let $(X_n; n \geq 1)$ be a sequence of random variables such that X_n has a binomial distribution with parameters n and λ/n, $\lambda > 0$. Then

$$\mathbf{E}(s^{X_n}) = \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n}s\right)^n \to e^{\lambda(s-1)} \quad \text{as } n \to \infty.$$

This is the p.g.f. of a Poisson random variable, and so as $n \to \infty$,

$$\mathbf{P}(X_n = k) \to e^{-\lambda}\lambda^k/k! \qquad \bullet$$

It is often convenient to work with distributions and moment generating functions. In this case the following result (for which we give no proof) is useful.

(3) Theorem Continuity *Let $\{F_n(x); n \geq 1\}$ be a sequence of distribution functions with corresponding moment generating functions $\{M_n(t) : n \geq 1\}$. If $F(x)$ is a distribution having corresponding moment generating function $M(t)$, then, as $n \to \infty$, $M_n(t) \to M(t)$ for all t, if and only if $F_n(x) \to F(x)$, whenever $F(x)$ is continuous.* $\bullet$

Additional conditions are required to link the convergence of a sequence of mass functions or distributions and the convergence of their moments. The following theorem (for which again we offer no proof) is for a sequence of distributions.

(4) Theorem *Suppose that for each $n \geq 1$, the distribution $F_n(x)$ has moments $\{\mu_j(n); j \geq 1\}$, such that $|\mu_j(n)| < a_j < \infty$.*

(i) Let $F_n(x) \to F(x)$, as $n \to \infty$, wherever $F(x)$ is continuous. Then as $n \to \infty$, for each j,

$$\mu_j(n) \to \mu_j < \infty,$$

and $(\mu_j; j \geq 1)$ are the moments of $F(x)$.

(ii) Conversely, for each $j \geq 1$, as $n \to \infty$, suppose that

$$\mu_j(n) \to \mu_j < \infty,$$

where $\{\mu_j; 1 \leq j\}$ are the moments of a unique distribution $F(x)$. Then, as $n \to \infty$, $F_n(x) \to F(x)$ wherever $F(x)$ is continuous.

There is a corresponding result for sequences of mass functions.

These theorems find applications (for example) in the theory of random graphs, and other combinatorial problems where moments are more tractable than distributions.

(5) Example Let X_n have the binomial distribution with parameters n and λ/n. Then, by Example 6.2.11,

$$\mu^{(k)} = \left\{ \begin{array}{ll} \frac{n!}{(n-k)!}\left(\frac{\lambda}{n}\right)^k & 1 \leq k \leq n \\ 0 & k > n \end{array} \right\} \to \lambda^k,$$

as $n \to \infty$. But, by Example 6.2.12, these are the factorial moments of the Poisson distribution. Hence, as $n \to \infty$, $\mathbf{P}(X_n = k) \to e^{-\lambda}\lambda^k/k!$, which we proved directly in Example 6.6.2, and earlier in Example 4.10. ●

(6) Example: Matching Again Recall that we are assigning n distinct letters randomly to n matching envelopes, and X is the number of matched pairs (of letter and envelope) that result. Consider the kth factorial moment of X.

$$\mu^{(k)} = \mathbf{E}(X(X-1)\ldots(X-k+1); \quad 1 \leq k \leq n.$$

Let I_j be the indicator of the event that the jth envelope contains the matching letter. Then the sum

$$S = \sum_{j_1 < j_2 < \ldots < j_k} I_{j_1} \ldots I_{j_k} \tag{7}$$

is just the number of ways of choosing a set of size k from the set of matching pairs. Another way of writing this is as $\binom{X}{k}$. Hence,

$$\begin{aligned} \frac{\mu^{(k)}}{k!} &= \mathbf{E}\frac{(X(X-1)\ldots(X-k+1))}{k!} \\ &= \mathbf{E}\left(\binom{X}{k}\right) \\ &= \mathbf{E}(S) \\ &= \binom{n}{k}\mathbf{E}(I_{j_1}\ldots I_{j_k}) \end{aligned} \tag{8}$$

$$= \binom{n}{k}\mathbf{P}\text{ (a given set of } k \text{ all match)}$$

$$= \binom{n}{k}\left(\frac{1}{n}\cdot\frac{1}{n-1}\cdots\frac{1}{n-k+1}\right) = \frac{1}{k!}.$$

Hence,

$$\mu^{(k)} = \left\{\begin{matrix} 1; & k \le n \\ 0; & k > n, \end{matrix}\right\} \to 1 \quad \text{for all } k$$

as $n \to \infty$. But these are the factorial moments of the Poisson distribution with parameter 1, and so as $n \to \infty$

$$\mathbf{P}(X = k) \to \frac{1}{ek!}. \tag{9}$$ ●

We conclude with an example that leads into the material of Chapter 7.

(10) Example 6.3.6 Revisited Recall that the p.g.f. of T_n (where T_n is the number of uniform random variables required to give a sum greater than n) is

$$G(s) = 1 + (s-1)\left(1 + \frac{s}{n}\right)^n.$$

What happens as $n \to \infty$? How do you interpret this?

Solution From (6.3.12), as $n \to \infty$,

$$\mathbf{E}(s^{T_n}) \to 1 + (s-1)e^s$$

or, equivalently,

$$\mathbf{E}(e^{tT_n}) \to 1 + (e^t - 1)e^{e^t}.$$

It follows that

$$\mathbf{P}(T_n = k) \to \frac{1}{(k-1)!} - \frac{1}{k!}.$$

The limiting factorial moments have a simple form for

$$\mathbf{E}((1+t)^{T_n}) \to 1 + te^{1+t} = 1 + \sum_{k=1}^{\infty}\frac{e}{(k-1)!}t^k.$$

Hence, in the limit $\mu^{(k)} = ek$, $k \ge 1$.

To interpret this, we return to the original definition

$$T_n = \min\left\{k : \sum_1^n X_i > n\right\} = \min\left\{k : \sum_1^n \left(\frac{X_i}{n}\right) > 1\right\},$$

where each X_i/n is uniformly distributed on $\{1/n, 2/n, \ldots, 1\}$. However, the limit of this sequence of uniform mass functions as $n \to \infty$ is *not* the mass function of a discrete random variable.

Intuitively, you may feel that it is approaching the distribution of a variable that is uniformly distributed over the *interval* [0, 1]. This vague remark can in fact be given a meaning if we introduce new objects – namely, continuous random variables. This is the subject of the next two chapters; after much technical development, it can be shown that the limit of T_n above is indeed the number of independent random variables, each uniform on [0, 1], required to produce a sum greater than 1.

6.7 Regeneration

Many interesting and important sequences of random variables arise as some process evolves in time. Often, a complete analysis of the process may be too difficult, and we seek simplifying ideas. One such concept, which recurs throughout probability, is the idea of regeneration. Here is an illustration:

(1) Example: Maze You are trying to traverse an unknown labyrinth. You set off at a constant speed from the clock by the portal, and each time a decision is required you choose at random from the alternatives. It is dark, and you have no pencil and paper, so a description of the process (i.e., your route) is impossible.

However, each time you arrive back at the portal, you can look at the clock and record T_r, the time at which you return to the clock for the rth time; $T_0 = 0$ say. Now it is clear from the setup that, when you set off for a second time (at time T_1), your chance of following any given route around the maze is the same as when you set off for the first time. Thus, the time until your second return, which is $T_2 - T_1$, has the same distribution as T_1 and is independent of T_1. The same is true of every subsequent interval between successive visits to the portal. These times $(T_n; n \geq 0)$ are regenerative in the sense that the distribution of your paths starting from T_n is the same as the distribution starting from T_m, for all m and n.

Of course, if you leave pebbles at junctions, or take a ball of string, or make a map, then this is no longer true. ●

Here is another archetypal illustration.

(2) Example: Renewal A machine started at $T_0 = 0$ uses a bit that wears out. As soon as it wears out, at time T_1, it is replaced by a similar bit, which in turn is replaced at T_2. Assuming that the machine performs much the same tasks as time passes, it seems reasonable to assume that the collection $(X_n; n \geq 1)$, where $X_n = T_n - T_{n-1}$, are independent and identically distributed. The replacement (or renewal) times are regenerative. ●

In fact, we have already used this idea of restarting from scratch; see, for example, Examples 4.4.9, 4.19, and 5.4.13.

Here is another elementary illustration.

Example Three players A, B, and C take turns rolling a fair die in the order $ABCAB\ldots$ until one of them rolls a 5 or a 6. Let X_0 be the duration of the game (i.e., the number of rolls). Let A_0 be the event that A wins, and let A_r be the event that A wins after the rth roll. Let W_i, $i \geq 1$ be the event that the game is won on the ith roll.

Of course,

$$A_0 = W_1 \cup \{W_1^c \cap W_2^c \cap W_3^c \cap A_3\}, \text{ where } \mathbf{P}(W_1) = \frac{1}{3}.$$

Now, as usual, we denote the indicator of any event E by $I(E)$, and so

$$\mathbf{E}(s^{X_0} I(W_1)) = \frac{1}{3}s.$$

Next we observe that if the first three rolls fail to yield 5 or 6, then the process regenerates (in the sense discussed above), so $X_0 = 3 + X_3$, where X_3 has the same distribution as X_0. Hence,

$$\mathbf{E}(s^{X_0} I(A_3)) = \mathbf{E}(s^{3+X_3} I(A_3)) = s^3 \mathbf{E}(s^{X_0} I(A_0)).$$

Therefore, we can write

$$\begin{aligned}\mathbf{E}(s^{X_0} I(A_0)) &= \mathbf{E}\big(s^{X_0}\big\{I(W_1) + I\big(W_1^c \cap W_2^c \cap W_3^c \cap A_3\big)\big\}\big) \\ &= \frac{1}{3}s + \frac{8}{27}s^3 \mathbf{E}(s^{X_0} I(A_0)).\end{aligned}$$

Hence,

$$\mathbf{E}(s^{X_0} I(A_0)) = \frac{\frac{1}{3}s}{1 - \frac{8}{27}s^3}.$$

Likewise, in an obvious notation,

$$\mathbf{E}(s^{X_0} I(B_0)) = \frac{\frac{2}{9}s^2}{1 - \frac{8}{27}s^3},$$

and

$$\mathbf{E}(s^{X_0} I(C_0)) = \frac{\frac{4}{27}s^3}{1 - \frac{8}{27}s^3}.$$

Hence,

$$\mathbf{E}(s^{X_0}) = \mathbf{E}(s^{X_0}(I(A_0) + I(B_0) + I(C_0))) = \frac{4s^3 + 6s^2 + 9s}{27 - 8s^3}. \qquad \bullet$$

Now we consider a more general case.

Let H be some phenomenon (or happening) that may occur or not at any time $n = 1, 2, 3, \ldots$. Let H_n be the event that H occurs at time n, and define X_n to be the time interval between the $(n-1)$th and nth occurrences of H. Thus,

$$X_1 = \min\{n > 0 : H_n \text{ occurs}\}$$

$$X_1 + X_2 = \min\{n > X_1 : H_n \text{ occurs}\}$$

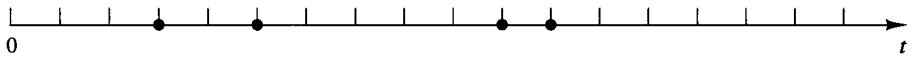


Figure 6.2 A delayed renewal process. Here, $X_1 = T_1 = 3; X_2 = 2, X_3 = 5, X_4 = 1, \ldots$

and so on. We suppose that $(X_n; n \geq 2)$ are independent and identically distributed random variables with mass function $(f(k); k \geq 1)$ and p.g.f. $G(s)$. The first interval X_1 is independent of $(X_n; n \geq 2)$, but its mass function may or may not be the same as that of X_2. This gives rise to two cases:

Case (O) The *ordinary* case. The mass function of X_1 is $(f(k); k \geq 1)$, the same as X_2.

Case (D) The *delayed* case. The mass functions of X_1 is $(d(k); k \geq 1)$, and $\mathbf{E}(s^{X_1}) = D(s)$.

These two cases admit a conventional interpretation: in the ordinary case, we suppose that H_0 occurred, so X_1 has the same mass function as the other intervals; in the delayed case, H_0 did not occur, so X_1 may have a different mass function.

The mathematical structure described above is known as a *recurrent event process*, or alternatively as a *discrete renewal process*. The important point about such a process is that each time H occurs, the process regenerates itself, in the sense discussed above. Figure 6.2 displays a realization of a renewal process.

Now Examples 1 and 2 make it clear that there are two essentially different types of renewal process. In Example 1, there is always a chance that you do traverse the maze (or encounter the Minotaur), and so do not return to the entrance. That is,

$$\mathbf{P}(X_2 < \infty) < 1.$$

Such a process is called *transient*. However, all machine bits wear out eventually, so in (2) we have

$$\mathbf{P}(X_2 < \infty) = 1.$$

Such a process is called *persistent* (or recurrent).

Now define the probabilities $u_n = \mathbf{P}(H_n)$, $n \geq 1$. A natural question is to ask whether this distinction between persistence and transience can also be observed in the properties of u_n. (The answer is yes, as we will see.)

It is customary to make a further distinction between two different types of persistent renewal process.

Definition *If* $\mathbf{E}(X_2) = \infty$, *then the process is said to be* null; *if* $\mathbf{E}(X_2) < \infty$, *then the process is said to be* nonnull.

Note that $\mathbf{E}(X_2)$ *is sometimes known as the* mean recurrence time *of the process.* ▲

Ordinary renewal is just a special case of delayed renewal, of course, but it is convenient to keep them separate. We therefore define $u_0 = 1$ and $u_n = \mathbf{P}(H_n)$, in the ordinary case; and $v_0 = 0$, and $v_n = \mathbf{P}(H_n)$, in the delayed case. These have respective generating

functions, $U(s) = \sum_0^\infty u_n s^n$ and $V(s) = \sum_1^\infty v_n s^n$. [Remember that $U(s)$ and $V(s)$ are not *probability* generating functions in the sense in which we use that term.]

Now we have the following:

(3) Theorem (i) $$U(s) = \frac{1}{1 - G(s)}$$

(ii) $$V(s) = \frac{D(s)}{1 - G(s)}.$$

Proof By conditional probability, in Case (O),

$$u_n = \sum_{k=1}^{n} \mathbf{P}(H_n | X_1 = k)\mathbf{P}(X_1 = k).$$

However, given H_k, the probability of any later occurrence of H is as if the process started at k. That is to say

$$\mathbf{P}(H_n | X_1 = k) = \mathbf{P}(H_n | H_k) = \mathbf{P}(H_{n-k}) = u_{n-k}, \quad n \geq k. \tag{4}$$

Hence,

$$u_n = \sum_{k=1}^{n} u_{n-k} f(k).$$

Because the right-hand sum is a convolution, its generating function is the product of the two generating functions $U(s)$ and $G(s)$, whence

$$U(s) - 1 = U(s)G(s). \tag{5}$$

Likewise, in Case (D),

$$\begin{aligned} v_n &= \sum_{k=1}^{n} \mathbf{P}(H_n | X_1 = k)\mathbf{P}(X_1 = k) = \sum_{k=1}^{n} \mathbf{P}(H_n | X_1 = k)\, d(k) \\ &= \sum_{k=1}^{n} u_{n-k} d(k) \quad \text{by (4).} \end{aligned}$$

Hence,

$$V(s) = U(s)D(s) = \frac{D(s)}{1 - G(s)} \quad \text{by (5).} \tag{6}$$

■

Thus, given $G(s)$ [and $D(s)$ in the delayed case] we can in principle find $\mathbf{P}(H_n)$, the probability that H occurs at time n, by expanding $U(s)$ in powers of s.

Conversely, given $V(s)$ [and $U(s)$ in the delayed case] we can find $G(s)$ and $D(s)$, and also decide whether the process is transient or persistent.

(7) Corollary If $U(1) < \infty$, then the process is transient. Otherwise, it is persistent.

Proof This follows immediately from Theorem 3(i). ■

(8) Example: Coincidences Suppose that a number c of independent simple symmetric random walks are started simultaneously from the origin. Let H be the "happening" that they are all at the origin, so H_{2n} is the event that all the c walks are at 0 on the $2n$th step. Show that H is persistent when $c = 2$, but H is transient for $c \geq 3$.

Solution For $c = 2$, we recall that

$$\begin{aligned}\mathbf{P}(H_{2n}) = (u_{2n})^2 &= \left(\frac{1}{4^n}\binom{2n}{n}\right)^2 \\ &= \left(1-\frac{1}{2}\right)^2\left(1-\frac{1}{4}\right)^2\cdots\left(1-\frac{1}{2n}\right)^2 \\ &> \left(1-\frac{1}{2}\right)^2\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\cdots\left(1-\frac{1}{2n-1}\right)\left(1-\frac{1}{2n}\right) \\ &= \frac{1}{4n} \quad \text{on successive cancellation.}\end{aligned}$$

Hence, H is persistent as $\sum_n \mathbf{P}(H_{2n})$ diverges.

For $c \geq 3$, we have similarly that

$$\begin{aligned}\mathbf{P}(H_{2n}) = \left(\frac{1}{4^n}\binom{2n}{n}\right)^c &= \left(\left(1-\frac{1}{2}\right)^2\left(1-\frac{1}{4}\right)^2\cdots\left(1-\frac{1}{2n}\right)^2\right)^{c/2} \\ &< \left(\frac{1}{2}.\frac{2}{3}.\frac{3}{4}.\frac{4}{5}\cdots\frac{2n-1}{2n}.\frac{2n}{2n+1}\right)^{c/2} = \left(\frac{1}{2n+1}\right)^{c/2}.\end{aligned}$$

Hence, H is transient as $\sum_n (1/(2n+1))^{c/2} < \infty$. ●

(9) Example: Stationary Renewal Let $X > 0$ have p.g.f. $G(s)$. Show that if $\mathbf{E}(X) < \infty$, then

$$H(s) = \frac{1}{\mathbf{E}(X)}\ \frac{1-G(s)}{1-s}$$

is the p.g.f. of a nonnegative random variable.

Now consider the delayed case of a recurrent event process in which $\mathbf{E}(s^{X_2}) = G(s)$ and $\mathbf{E}(s^{X_1}) = H(s)$. Show that for all n

$$\mathbf{P}(H_n) = \frac{1}{G'(1)}. \tag{10}$$

Solution From (6.1.6), we have that $H(s)$ is a power series with nonnegative coefficients. Furthermore, by L'Hôpital's rule,

$$H(1) = \lim_{s\uparrow 1}\frac{-G'(s)}{-\mathbf{E}(X)} = 1$$

Hence, $H(s)$ is a p.g.f. Finally, if $D(s) = H(s)$ in (6), then

$$V(s) = \frac{1}{\mathbf{E}(X_2)(1-s)},$$

and the result follows. ●

6.8 Random Walks

Recall that if $(X_i; i \geq 1)$ are independent and identically distributed, then $S_n = S_0 + \sum_1^n X_i$ is a random walk. Because generating functions have been so useful in handling sums of random variables, we may expect them to be exceptionally useful in analysing random walks. If X has p.g.f. $G(z)$, then trivially we have, when $S_0 = 0$.

$$G_n(z) = \mathbf{E}(z^{S_n}) = (G(z))^n.$$

It follows that we can define the function H by

$$H(z, w) = \sum_{n=0}^{\infty} w^n G_n(z) = (1 - wG(z))^{-1}. \tag{1}$$

This bivariate generating function tells us everything about S_n in principle, as $\mathbf{P}(S_n = r)$ is the coefficient of $z^r w^n$ in $H(z, w)$. However, the analytical effort required to work at this level of generality is beyond our scope. We proceed by considering simple examples.

(2) Example: Simple Symmetric Random Walk Let $(S_n = \sum_1^n X_i; n \geq 0)$ be a simple symmetric random walk, with $S_0 = 0$. Let H_n be the event that $S_n = 0$. Because steps of the walk are independent and identically distributed, it follows that visits to the origin form an ordinary renewal process. Here, $u_n = \mathbf{P}(S_n = 0)$.

Define the first passage times,

$$T_j = \min\{n > 0 : S_n = j | S_0 = 0\},$$

and the generating functions, $U(s) = \sum_0^\infty u_n s^n$ and $G_j(s) = \mathbf{E}(s^{T_j})$.

Find $U(s)$ and $G_0(s)$, and show that the simple symmetric random walk is persistent null.

Solution We give two methods of finding $U(s)$ and $G_j(s)$. For the first, define

$$T_j^* = \min\{n: S_n = j - 1 | S_0 = -1\}$$

and let $\hat{T}_1$ be a random variable having the same distribution as T_1, but independent of T_1. Because the steps of the walk are independent, and symmetrically and identically distributed, it follows that

$$\mathbf{E}(s^{T_1}) = \mathbf{E}(s^{T_{-1}}), \tag{3}$$

$$T_2 = T_1 + \hat{T}_1, \tag{4}$$

and

$$\mathbf{E}(s^{T_2^*}) = \mathbf{E}(s^{T_2}) = \mathbf{E}(s^{T_1 + \hat{T}_1}) \tag{5}$$

$$= (G_1(s))^2, \quad \text{by independence.} \tag{6}$$

Hence, by conditional expectation,

$$G_1(s) = \mathbf{E}(\mathbf{E}(s^{T_1}|X_1)) = \frac{1}{2}\mathbf{E}(s^{T_1}|X_1 = 1) + \frac{1}{2}\mathbf{E}(s^{T_1}|X_1 = -1)$$
$$= \frac{1}{2}s + \frac{1}{2}\mathbf{E}(s^{1+T_2^*}) = \frac{1}{2}s + \frac{1}{2}s(G_1(s))^2 \quad \text{by (6).} \tag{7}$$

One root of (7) is a probability generating function, so this root is $G_1(s)$, namely,

$$G_1(s) = (1 - (1 - s^2)^{\frac{1}{2}})/s.$$

Now, using conditional expectation again,

$$\begin{aligned} G_0(s) &= \mathbf{E}(\mathbf{E}(s^{T_0}|X_1)) \\ &= \frac{1}{2}\mathbf{E}(s^{T_0}|X_1 = 1) + \frac{1}{2}\mathbf{E}(s^{T_0}|X_1 = -1) \\ &= \frac{1}{2}s\mathbf{E}(s^{T_1}) + \frac{1}{2}s\mathbf{E}(s^{T_{-1}}) = sG_1(s) \quad \text{by (3)} \\ &= 1 - (1 - s^2)^{\frac{1}{2}}. \end{aligned} \tag{8}$$

Hence,

$$U(s) = (1 - s^2)^{-1/2} \quad \text{by Theorem 6.7.3.} \tag{9}$$

Alternatively, we could observe that $S_{2n} = 0$ if and only if the walk has taken n steps in each direction. They may be taken in any order so

$$u_{2n} = \binom{2n}{n}\left(\frac{1}{2}\right)^{2n}. \tag{10}$$

Now recall that by the negative binomial theorem

$$\sum_0^\infty \binom{2n}{n}\left(\frac{1}{2}\right)^{2n} x^n = (1 - x)^{\frac{-1}{2}}, \tag{11}$$

and (9) and (8) follow.

Setting $s = 1$ shows that $G_0(1) = 1$ (and $U(1) = \infty$) so H is persistent. However,

$$\frac{d}{ds}G_0(s) = \frac{s}{(1 - s^2)^{\frac{1}{2}}},$$

and setting $s = 1$ shows that H is null; the expected number of steps to return to the origin is infinite as we know already, recall Example 5.6.27. ●

Now that we have the generating functions $U(s)$ and $G_0(s)$, we can provide slicker derivations of earlier results. For example,

$$G_0(s) = 1 - (1 - s^2)^{\frac{1}{2}} = 1 - (1 - s^2)U(s).$$

Hence, equating coefficients of s^{2k} gives (5.18.1)

$$f_{2k} = u_{2k-2} - u_{2k}. \tag{12}$$

Also,

$$s\frac{d}{ds}G_0(s) = s^2(1-s^2)^{-\frac{1}{2}} = s^2U(s)$$

and so equating coefficients again gives (5.18.6)

$$2kf_{2k} = u_{2k-2}. \tag{13}$$

See Problem 41 for another simple application of this. Here is a trickier application.

(14) Example: Truncated Walk Let $(S_n; n \geq 1)$ be a simple symmetric random walk with $S_0 = 0$, and let $T = \min\{n > 0 : S_n = 0\}$. Let $T \wedge 2m = \min\{T, 2m\}$ and show that

$$\mathbf{E}(T \wedge 2m) = 4mu_{2m} = 2\mathbf{E}(|S_{2m}|). \tag{15}$$

Solution We establish (15) by showing that all three terms have the same generating function. Equality then follows by the uniqueness theorem. First,

$$\sum_0^\infty 4mu_{2m}s^{2m} = 2s\sum_1^\infty 2ms^{2m-1}u_{2m} = 2s\frac{d}{ds}U(s) = \frac{2s^2}{(1-s^2)^{3/2}}.$$

Second, recalling (13) and (5.18.2),

$$\mathbf{E}(T \wedge 2m) = \sum_{k=1}^m 2kf_{2k} + 2m\mathbf{P}(T > 2m) = \sum_{k=1}^m u_{2k-2} + 2mu_{2m}.$$

Hence,

$$\begin{aligned}\sum s^{2m}\mathbf{E}(T \wedge 2m) &= \sum_m s^{2m}\sum_{k=0}^{m-1} u_{2k} + s\sum_m 2ms^{2m-1}u_{2m} \\ &= \frac{s^2U(s)}{1-s^2} + sU'(s) \quad \text{using (6.1.10)} \\ &= \frac{2s^2}{(1-s^2)^{\frac{3}{2}}}.\end{aligned}$$

Finally, using the hitting time theorem (5.6.17),

$$\begin{aligned}\sum_m s^{2m}\mathbf{E}(|S_{2m}|) &= 2\sum_m s^{2m}\sum_{k=1}^m 2k\mathbf{P}(S_{2m} = 2k) \\ &= 2\sum_m s^{2m}\sum_{k=1}^m 2mf_{2k}(2m) \\ &= 2s\frac{d}{ds}\sum_m s^{2m}\sum_{k=1}^m f_{2k}(2m)\end{aligned}$$

$$= 2s\frac{d}{ds}(G_2(s) + G_4(s) + G_6(s) + \ldots)$$
$$= 2s\frac{d}{ds}\frac{(G_1(s))^2}{(1-(G_1(s))^2)} \quad \text{by Example (2)}$$
$$= s\frac{d}{ds}((1-s^2)^{-\frac{1}{2}} - 1) = \frac{s^2}{(1-s^2)^{\frac{3}{2}}}. \qquad \bullet$$

As a final example of the use of generating functions in random walks, we establish yet another arc-sine law.

(16) Example: Arc-Sine Law for Leads Let $(S_n; n \geq 0)$ be a simple symmetric random walk with $S_0 = 0$. Of the first $2n$ steps, let L_{2n} denote the number that do not enter the negative half-line. Show that

$$\mathbf{P}(L_{2n} = 2k) = 4^{-n}\binom{2k}{k}\binom{2n-2k}{n-k}. \tag{17}$$

Solution Define the generating functions

$$G_{2n}(s) = \mathbf{E}(s^{L_{2n}}) \qquad \text{and} \qquad H(s,t) = \sum_{n=0}^{\infty} t^{2n} G_{2n}(s). \tag{18}$$

Let T be the number of steps until the walk first revisits zero, and recall that

$$F(s) = \mathbf{E}(s^T) = \sum_{1}^{\infty} s^{2r} f(2r) = 1 - (1-s^2)^{\frac{1}{2}}. \tag{19}$$

Now using conditional expectation

$$\mathbf{E}(s^{L_{2n}}) = (\mathbf{E}(\mathbf{E}(s^{L_{2n}}|T)$$
$$= \sum_{r=1}^{n} \mathbf{E}(s^{L_{2n}}|T = 2r) f(2r) + \mathbf{E}(s^{L_{2n}}|T > 2n) \sum_{r=n+1}^{\infty} f(2r).$$

Now, depending on the first step,

$$\mathbf{P}(L_T = T) = \frac{1}{2} = \mathbf{P}(L_T = 0),$$

and visits to zero constitute regeneration points for the process L_{2n}. Hence, we may rewrite (18) as

$$G_{2n}(s) = \sum_{r=1}^{n} G_{2n-2r}\left(\frac{1}{2}(s^{2r}+1)\right) f(2r) + \sum_{r=n+1}^{\infty} f(2r)\left(\frac{1}{2}(s^{2n}+1)\right).$$

Multiplying by t^{2n} and summing over n yields

$$H(s,t) = \frac{1}{2}H(s,t)(F(st) + F(t)) + \frac{1}{2}.\frac{1-F(st)}{1-t^2s^2} + \frac{1}{2}.\frac{1-F(t)}{1-t^2},$$

by the convolution theorem. Now substituting for $F(.)$ from (19) gives

$$H(s,t) = ((1-s^2t^2)(1-t^2))^{-\frac{1}{2}}.$$

The coefficient of $t^{2n}s^{2k}$ in this is (17). Now use Exercise 5.18.8 to produce the arc-sine distribution. ●

6.9 Review and Checklist for Chapter 6

All random variables have a probability distribution, and many of them also have moments. In this chapter, we introduced two miraculous devices to help us with many of the chores involved in handling and using probabilities and moments.

Probability generating function of X: p.g.f.

$$G_X(s) = \sum_n \mathbf{P}(X = n)s^n = \mathbf{E}s^X.$$

Moment generating function of X: m.g.f.

$$M_X(t) = \sum_n \mathbf{P}(X = n)e^{nt} = \mathbf{E}e^{tX}.$$

You can think of these functions as organizers that store a collection of objects that they will regurgitate on demand. Remarkably, they will often produce other information if suitably stimulated; thus, the p.g.f. will produce the moments (if any exist), and the m.g.f. will produce the probability distribution (in most cases). We used them to study sums of independent random variables, branching processes, renewal theory, random walks, and limits. They have these properties:

Connections:

$$M_X(t) = G_X(e^t) \quad \text{and} \quad G_X(s) = M_X(\log s).$$

Tails:

$$T(s) = \sum_{n=0}^{\infty} s^n \mathbf{P}(X > n) = \frac{1 - G_X(s)}{1 - s}, \quad \text{when } X \geq 0.$$

Uniqueness:

$$f_X(k) = G_X^{(k)}(0)/k!.$$

Moments:

$$\mu^{(k)} = G_X^{(k)}(1) \quad \text{and} \quad \mathbf{E}X^k = M_X^{(k)}(0)$$
$$\mu^{(k)} = kT^{(k-1)}(1).$$

[Where $\mu^{(k)}$ is the kth factorial moment.]

Sums and random sums: For independent $(X_n; n \geq 1)$,

$$\mathbf{E}s^{X_1+\cdots X_n} = \prod_{r=1}^{n} G_{X_r}(s); \quad \mathbf{E}\exp(t(X_1 + \cdots X_n)) = \prod_{r=1}^{n} M_{X_r}(t).$$

For independent $(X_n; n \geq 1)$ and independent nonnegative integer valued N,

$$\mathbf{E}s^{X_1+\cdots X_N} = G_N(G_X(s)); \quad \mathbf{E}\exp(t(X_1 + \cdots + X_N)) = G_N(M_X(t)).$$

Joint generating functions:

$$G_{\mathbf{X}}(\mathbf{s}) = \mathbf{E}\prod_{r=1}^{n} s_r^{X_r}$$

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbf{E}\left[\exp\left(\sum_{r=1}^{n} t_r X_r\right)\right]$$

Independence: X and Y are independent if and only if

$$\mathbf{E}(x^X y^Y) = G_X(x)G_Y(y), \quad \text{for all } x, y, \text{ or}$$

$$\mathbf{E}[\exp(sX + tY)] = M_X(s)M_Y(t), \quad \text{for all } s, t.$$

Branching: If the family size p.g.f. is $G(s)$, then

$$G_{m+n}(s) = \mathbf{E}s^{Z_{m+n}} = G_m(G_n(s)) = G_n(G_m(s)).$$

The probability η of ultimate extinction is the smallest positive root of $G(x) = x$.

Special generating functions:

Binomial distribution $(q + ps)^n$
Uniform distribution on $\{0, 1, \ldots, n\}$ $\quad \frac{1-s^{n+1}}{(n+1)(1-s)}$
Geometric distribution $\frac{ps}{1-qs}$
Poisson distribution $e^{\lambda(s-1)}$
Negative binomial distribution $(\frac{ps}{1-qs})^n$
Logarithmic distribution $\frac{\log(1-sp)}{\log(1-p)}$

Checklist of Terms for Chapter 6

6.1 probability generating function
tail generating function
6.2 uniqueness theorem
factorial moments
6.3 sums and random sums
branching process
extinction probability
6.4 moment generating function
cumulant generating function
6.5 joint probability generating function
factorization and independence
6.6 continuity theorem
6.7 renewal process
persistent
transient
null, nonnull
6.8 simple random walk
arc-sine law for leads